Canadian Community Health Survey (CCHS)

Annual component

User guide

2023 Microdata file

October 2024

Statistics Canada Statistique Canada

Canada

TABLE OF CONTENT

1. INTRODUCTION 8
2. BACKGROUND 9
3. CCHS REDESIGNS 10
3.1 2015 CCHS Redesign 10
3.2. 2022 CCHS Redesign 10
4. CONTENT STRUCTURE OF THE CCHS 11
4.1 Core content 11
4.2 Theme content 11
4.3 Optional content 11
4.4 Rapid response content 11
4.5 Content included in data files 12
5. SAMPLE DESIGN 12
5.1 Target population 12
5.2 Health regions 12
5.3 Frames, household sampling strategies 12
5.3.1 Sampling of households from the area frame 13
5.3.2 Sampling of households from the 2021 Census of population for the DDAP initiative 14
5.4 Sample size and allocation 15
5.5 Sample allocation over the collection periods 15
5.6 Sampling of interviewees 16
6. DATA COLLECTION 16
6.1 Electronic questionnaires 16
6.2 CCHS application development 17
6.3 Interviewer training 18
6.4 The interview 18
6.5 Field operations 20
6.6 Quality control and collection management 20
7. DATA PROCESSING 20
7.1 Editing 20
7.2 Flows 21
7.3 Coding 22
7.4 Creation of derived variables 23
7.5 Weighting 23
7.6 Income variables for 2023 - Linkage and Imputation 23
7.7 Postal code imputation 24
7.8 Gender coding 24
8. WEIGHTING 25
8.1 Overview 25
8.2 Weighting of the area frame sample 27
8.3 Weighting of the DDAP frame sample 29
8.4 Final weight adjustments with area and DDAP frames combined 30
8.5 Creation of a share weight 32
9. DATA QUALITY 33
9.1 Response rates 33
9.2 Survey Errors 33
9.2.1 Non-Sampling Errors 34
9.2.2 Sampling Errors 34
10. GUIDELINES FOR TABULATION, ANALYSIS AND RELEASE 35
10.1 Rounding guidelines 35

10.2 Sample weighting guidelines for tabulation 36
10.2.1 Definitions: categorical estimates, quantitative estimates 36
10.2.2 Tabulation of categorical estimates 37
10.2.3 Tabulation of quantitative estimates 37
10.3 Guidelines for statistical analysis 37
10.4 Release guidelines 38
10.4.1 Master or share file 39
10.4.2 Release rules for estimated proportions and estimated counts 40
10.4.3 Release rules for estimated counts, estimated means and estimated totals 40
10.4.4 Release rules for differences and ratios 42
10.4.5 Additional Rules Regarding Confidence intervals 42
11. APPROXIMATE SAMPLING VARIABILITY TABLES 42
11.1 How to use the CV tables to obtain confidence limits 45
11.2 How to use the CV tables for categorical estimates 46
11.3 Examples of using the CV tables for categorical estimates to calculate confidence intervals 48
11.4 How to use the CV tables to do a Z-test 51
11.5 Example of using the CV tables to do a Z-test 51
11.6 Exact variances or coefficients of variation 52
12. MICRODATA FILES: DESCRIPTION, ACCESS AND USE 53
12.1 Master files 53
12.1.1 Research Data Centre 53
12.1.2 Custom tabulations 53
12.1.3 Remote access 53
12.1.4 Real Time Remote Access (RTRA) 54
12.2 Share files 54
12.3 Public use microdata files 54
12.4 How to use the CCHS data files: annual data file or two-year data file? 55
12.5 Use of weight variable 56
12.6 Variable naming convention 56
12.7 Data dictionaries 58
APPENDIX A - CANADIAN COMMUNITY HEALTH SURVEY (CCHS) - CONTENT OVERVIEW – 2023 59
APPENDIX B - AVAILABLE GEOGRAPHY IN THE MASTER AND SHARE FILES AND THEIR CORRESPONDING CODES: CANADA, PROVINCES, TERRITORIES, HEALTH REGIONS AND PEER GROUPS 62
APPENDIX C - SAMPLE ALLOCATION BY PROVINCE (CCHS 2023) 67
APPENDIX D - RESPONSE RATES BY PROVINCE (CCHS 2023) 69

WHAT'S NEW IN THE 2023 CANADIAN COMMUNITY HEALTH SURVEY (CCHS)?

In addition to the previous 2015 redesign, the CCHS underwent a second major redesign in 2022. This redesign centered on a collection mode change from computer-assisted telephone interview (CATI)/computer-assisted personal interview (CAPI) to an electronic questionnaire (EQ) format with CATI and CAPI follow-up for non-response. As a result, caution should be used when comparing data between various cycles (i.e., pre-2015 data, 2015-2021 data, 2022-present data).

Additionally, the sample size and coverage for the Territories was increased in order to have full coverage of the North for the 2022 file. Since the Territories are normally covered only by two-year files, the data for the North for 2022 is kept on a separate microdata file to continue having the annual file cover only the provinces. Also, the sampling plan for the North has returned to the two-year design for 2023-2024, so annual coverage of the North in 2022 is a unique occurrence this year.

Sampling

For the 2023 cycle of CCHS, the coverage of the Canadian population by age was adjusted to cover only the adult population aged 18 and older, dropping the coverage of those aged 12 to 17. This change implemented because of the launch of the Canadian Health Survey on Children and Youth, which will cover the youth aged 12 to 17 with a richer sample size and content more appropriately designed to cover the health characteristics, behaviours, and outcomes for this population.

Census 2021 was chosen as an additional frame for the CCHS 2023 as it is the best option to allow stratification by population characteristics. Census 2021 data was used to target households with members who identified on the Census 2021 as belonging to groups that are part of the Disaggregated Data Action Plan (DDAP) initiative. An extra sample of 37,000 households was selected from Census 2021 long form data. These households were allocated equally across the last three collection periods of 2023. This extra sample was selected at the province and equity group level.

Collection

In 2023, CCHS collection was updated to enable respondents to answer the survey directly via an online electronic questionnaire (EQ) with computer-assisted telephone interview (CATI) and computer-assisted personal interview (CAPI) follow-up for non-response. Prior to 2022, CCHS content was designed exclusively for computer-assisted interviewing (CAI). The transition to EQ collection was a significant change that required substantial format updates to all CCHS

modules. For this reason, caution should be used when comparing data between various cycles (i.e., pre 2015 data, 2015-2021 data, data released for the 2022 cycle onwards.)

Content

As mentioned above, the transition to EQ collection, required many format updates to all CCHS modules. For example, the following changes were applied to all CCHS content where applicable.

- Interviewer notes for probing respondents were replaced with condensed on-screen help text.
- Questions and corresponding answer categories were shortened for clarity.
- Repetitive questions were combined under one header rather than having a series of individual questions (e.g., using a grid to ask multiple related questions).
- The number of edits to verify respondent's answers was reduced.
- Other than questions required to verify the respondent's identity, non-response was allowed for all content (i.e., the ability to skip questions).

The following represents a high-level overview of the updates to CCHS Core and Theme content:

Changes to core modules:

- A new stand-alone Date of birth (AGE) module; this module was previously a sub-block of the **Age of respondent module (ANC1).**
- The **Chronic conditions (CCC)** module was split into two modules: an updated CCC module and a new **Chronic conditions 1 (CC1)** module.
- **Smoking (SMK)** was divided into three new modules: **Current smoking status (CSS)**; **Smoking – past use (SPU)**; and **Electronic cigarettes and vaping (ECV)**.
- Life satisfaction measures (LSM) was added as a new sub-block of General health (GEN). Previously, this module was part of the Subjective well-being minimum (SLM) sub-block.
- A new stand-alone Pregnancy (PRS) module; this module was previously part of the Main activity (MAC) module.
- A new stand-alone Perception of weight (WTP) module; this module was previously part of the Height and weight (HWT).
- The Relationship matrix (REL) module and Relationship without confirmation (RNC) sub-block were combined to create the new Relationship with confirmation (RWC) module.
- The following modules were updated to align with Statistics Canada's standard Harmonized Content:
    - Main activity module (MA) and School attendance (EDC) sub-block
    - Place of birth, immigration, and citizenship (IM)
    - Indigenous identity (ABM)
    - Sexual orientation (SOR)

- Verification (VER2).
- The Electronic cigarettes and vaping 2 (ECV2) module no longer exists since the questions are now in the Electronic cigarettes and vaping (ECV) module.
- A new stand-alone Long-term condition (LTC) module; this module is on identifying as a person with a disability.
- The Main Activity 1 (MAC) module no longer exists because it is now part of the Main Activity 2 (MA) module.
- Food security (FSC) was a 2-year theme in 2021- 2022 but added to the core content in 2023.
- Home care services (HMC) was 2-year theme in 2021-2022 but added to the core content in 2023.
- Regular health care provider (RHC) is now a core module (was a theme module in 2022)

Changes to theme modules

- The Eating Habits (EAH) module was redesigned based on the original Fruit and vegetable consumption (FVC) module.
- Flu shots (FLU) was added as a theme module in 2022; previously it was part of core CCHS content from 2015-2020.
- Oral health (OHM) and Oral health 3 (OHM3) were introduced in 2022 to collect data on oral health and regular dental care habits (modified from the optional module OHT in 2020).
- Prescription cost (PCN) was added to collect data on non-adherence to prescription medicine due to cost (modified from the 2016 rapid response module Prescription - Cost-related non-adherence).
- Medication use – pain relievers (PRM) was added as a stand-alone module. It was previously part of the Medication use (MED) module and collects information on the use of prescription and non-prescription pain relief medications. These questions were previously part of theme content in 2018 and optional content in 2019 and 2020.
- Vaccination passport COVID-19 (PVC) was added to collect data on COVID-19 vaccination passports.
- Regular health care provider (RHC) was added to collect data on whether respondents have a regular health care provider (modified from the previous Primary health care (PHC) module).
- Use of protective equipment (UPE) questions were re-formatted into a grid presentation. This was a theme module in 2017 and 2018 and an optional module in 2020.
- The module Washington Group – disability measure (WDM) was reformatted into a grid presentation. This was theme content in 2017 and 2018.
- Work family balance (WLB) and Work productivity (WPL) were added as new rapid response modules for 2022.

- **The COVID-19 2 (COV2)** module has been separated in seven small modules by subject: **COVA** (mental health during the pandemic), **COVB** (precautions taken during the pandemic), **COVC** (experienced symptoms of COVID-19), **COVD** (vaccination status), **COVE** (reasons for the vaccination status), **COVF** (likelihood of receiving a booster dose), **COVG** (likelihood of receiving an annual booster).
- **Abilities (WDM)** has been replaced by **Health utility index (HUI)**

New **Theme** modules for 2023:

- **Moderate physical activity (MPA)**
- **Other physical activity (OPA)**
- **Alcohol use during the past week (ALW)**

New **Optional** modules for 2023:

- **Use of protective equipment (UPE)** – was a 2-year theme module for 2021-2022
- **Activities of daily living (ADL)**
- **Maternal experiences (MEX)** – was a 2-year theme module for 2021-2022
- **Positive mental health (PMS)**
- **Consultations about mental health (CMH)**
- **Mental health (DEP)**
- **Social provisions (SPS)**

The following modules have been dropped from the survey for 2023:

- **Weight perception (WTP)**
- **Vaccination passport (PVC)**
- **Injuries (INJ)**
- **Oral Health (OHM and OHM3)**
- **Sedentary behaviours (SBE)**
- **Sleep (SLP)**
- **Maternal experiences – smoking during pregnancy (MXS)**
- **Maternal experiences – alcohol use during pregnancy (MXA)**
- **Insurance coverage (INL and INP)**
- **Prescription cost (PCN)**

For the adult sample, the CCHS returned to using the Labour Force Survey (LFS) area frame in 2022 instead of the Dwelling Universe File which was used in 2021.

For more detailed information on CCHS content, please see Appendix A.

Geography

The sampling for CCHS was implemented at the Health Region level using the HR definitions from 2018. For the purposes of weighting and dissemination, the included geography variable GEODVHR4 is based on the 2023 health region definitions. See (Health Regions: Boundaries and Correspondence with Census Geography (statcan.gc.ca)

Documentation

- Errata: Errors that are discovered in the CCHS data files and products after dissemination are communicated to users in a cumulative error log called "CCHS Errata". The errata provides details on:
    - the products affected (e.g. master or share files or PUMF);
    - the years affected;
    - suggested corrections (if possible); and
    - steps to carry out to apply the correction

Since the release of the 2021 CCHS, there has been one additional entry to the Errata document:

- Error in PHC_030 to PHC_060

For your convenience, the errata document has an index on the first page. It lists modules affected and the years affected. To see if a particular variable is affected, follow the hyperlinks on the associated errata item numbers.

An up to date copy of the CCHS Errata can be obtained by contacting the Health Statistics Branch Centre for Population Health Data at 613-951-1746 or HD-DS@statcan.gc.ca.

1. INTRODUCTION

The Canadian Community Health Survey (CCHS) is a cross-sectional survey that collects information related to health status, health care utilization and health determinants for the Canadian population. It surveys a large sample of respondents and is designed to provide reliable estimates at the health region level. Prior to 2022, the CCHS was administered through computer assisted interviewing, by telephone (CATI) or in person (CAPI). In 2022 the survey was transitioned to direct self-responding via an online electronic questionnaire, with the addition of computer assisted telephone interviews (CATI) and computer assisted personal interviews (CAPI) for non-responses. The CCHS had also undergone a major redesign that started in 2012 and was implemented in collection in 2015. A second redesign occurred in 2022. Details of these redesigns are provided in Chapter 3.

The survey's objectives are as follows:

- support health surveillance programs by providing health data at the national, provincial and health region levels;
- provide a single data source for health research on small populations and rare characteristics;
- timely release of information easily accessible to a diverse community of users; and
- create a flexible survey instrument that includes a rapid response option to address emerging issues related to the health of the population.

The CCHS data is normally collected from persons aged 18 and over living in private dwellings in over 100 health regions covering all provinces and territories. Excluded from the sampling frame are individuals living on Indian Reserves and on Crown Lands, institutional residents (health institutions, prisons, religious institutions, convents, etc.), full-time members of the Canadian Forces, and residents of certain remote regions. The CCHS covers approximately 97% of the Canadian population aged 18 and over.

The purpose of this document is to facilitate the manipulation of the CCHS microdata files and to describe the methodology used. The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). The characteristics of each of these files are presented in this guide. The master files and share files will be released on October 2nd, 2024.
Any questions about the data sets or their use should be directed to:

Survey Help Line:	1-800-949-9491
For custom tabulations or general data support:	
Client Custom Services, Health Statistics Branch:	HD-DS@statcan.ca
For remote access support:	statcan.cchssm-esccds.statcan@statcan.gc.ca

2. BACKGROUND

In 1991, the National Task Force on Health Information cited a number of issues and problems with the health information system. The members felt that data was fragmented; incomplete, could not be easily shared, was not being analysed to the fullest extent, and the results of research were not consistently reaching Canadians.[1]

In responding to these issues, the Canadian Institute for Health Information (CIHI), Statistics Canada and Health Canada joined forces to create a Health Information Roadmap in 1999. From this mandate, the Canadian Community Health Survey (CCHS) was conceived. The format, content and objectives of the CCHS evolved through extensive consultation with key experts and federal, provincial and community health region stakeholders to determine their data requirements.[2] The survey started data collection in 2000.

To meet many data requirements, the CCHS had a two-year data collection cycle. Up to 2007, the first year of the survey cycle, designated by ".1", was a general population health survey, designed to provide reliable estimates at the health region level. The second year of the survey cycle, designated by ".2", had a smaller sample and was designed to provide provincial level results focused on specific health topics.

Starting in 2007, the regional component of the CCHS program began collection on an ongoing basis. The x.1 cycles of the CCHS were renamed "the annual component" of the CCHS and are now collected every year. The full title of a given annual cycle is "The Canadian Community Health Survey – Annual component, 20XX" and the short title is simply "CCHS –20XX".

The focused content component of the survey has been designated by the name of the survey followed by the topic of the themes covered by each survey (e.g., “Canadian Community Health Survey on Healthy Aging” or “CCHS – Healthy Aging”).

In 2012, a major redesign began which changed the survey sampling and collection strategy. Survey content was qualitatively tested to ensure validity and reliability in preparation for a new 8 year content plan beginning in 2015.

In 2022, the CCHS underwent a second major redesign.

---

[1] Health Canada; Statistics Canada. 1999. *Health Information Roadmap: Responding to Needs.* p. 3.
[2] Canadian Institute for Health Information; Statistics Canada. 1999. *Health Information Roadmap: Beginning the Journey*. ISBN 1-895581-70-2. p. 19.

3. CCHS REDESIGNS

Below are the main changes resulting from the CCHS redesigns:

3.1 2015 CCHS Redesign

- Content changes were made to most modules, but mainly to the core and theme modules. Roughly 70% of the modules that existed before the redesign have seen some level of change. Those changes vary, and have involved either minor tweaks or major changes to concepts, vocabulary, or response categories. Some new modules were created to allow for emerging concepts or data gaps identified by data sharing partners such as Health Canada, the Public Health Agency of Canada and the provincial and territorial ministries of health.

- Variable conventions have changed in 2015. The naming convention used for variables in the 2015 CCHS use a maximum of eight characters. As before, positions 1 to 3 make up the module or questionnaire section name, while position 4 indicates the variable type (underscore, D, F, C or G). Survey elements in the questionnaire (e.g. E, C, D, Q, N) now follow a uniform convention that increases in increments of five (e.g. CCC_005, CCC_010, CCC_015).

- All the derived variables were reviewed, and specifications were updated with the new variable names. Content and conceptual changes were also reviewed, and new derived variables were created for new content modules. The annual data collection strategy changed for 2015. Prior to 2015, there were six two-month collection periods. Starting in 2015, survey collection was divided into four non-overlapping three month periods.

- Starting in 2015, changes to data collection in the North resulted in approximately half of the communities in each territory visited annually. As a result, the data is only representative in the territories after two years.

3.2. 2022 CCHS Redesign

The 2022 CCHS redesign resulted in major methodological and content changes.

Methodological changes:

- Prior to 2022, the CCHS was administered through computer assisted interviewing, by telephone (CATI) or in person (CAPI). In 2022 the survey was transitioned to direct self-response via an online electronic questionnaire (EQ), with the addition of computer assisted telephone interviews (CATI) and computer assisted personal interviews (CAPI) for non-response. Due to this transition to EQ format, a substantial number of modules have undergone changes. There is also the potential for collection mode effect with the transition to EQ, given that in 2021 all collection was done by telephone, while in 2022, 51% of cases were self-reported and completed online, 48.5% were completed by phone, and only 0.5% were completed in person. The response rate also varied widely between 2021 and 2022 (24.1% vs 43.0%).

- The CCHS transitioned from using Blaise software for computer-assisted interviewing and survey processing to an in-house Collection Management Portal (CMP).

Content changes in 2022:
- A significant number of variables changed names (219 out of 413 variables).
- Many response options changed (i.e., new, removed, or modified categories).
- Some questions were reordered within their modules and some text changed slightly or completely.

For more information on the CCHS 2023 content, please refer to Appendix A.

4. CONTENT STRUCTURE OF THE CCHS

In addition to sociodemographic and administrative data, the CCHS includes three content components, each of which addresses a different need: Common content (Core content and Theme content) and Rapid Response content. Appendix A lists the modules included in the 2023 questionnaire and their type of component.

4.1 Core content

The CCHS core content component includes questions asked of respondents in all provinces and territories (unless otherwise specified).

4.2 Theme content

The theme content is asked of all survey respondents and comprises modules, or in some cases groups of questions within modules, that are related to a specific topic. There are two types of themes collected: two-year theme modules which are asked over a period of two years and one-year theme modules that are asked over a one-year period. The one-year theme changes annually. Themes are reintroduced in the survey as required. This component enables CCHS to better plan its content in the medium term.

4.3 Optional content

Optional content includes modules that were selected by provinces and territories to meet their specific data needs.

4.4 Rapid response content

The rapid response component is offered on a cost recovery basis to organizations interested in obtaining national estimates on an emerging or specific topic related to the health of the population. The rapid response content takes a maximum of two minutes of interview time. The questions appear in the questionnaire for one or two collection periods (three to six months) and are asked of all CCHS respondents during that period, excluding the Territories.

In 2023, there were two rapid responses released together as a single file on November 6th 2023; Smoking – stage of change 2 (SCH2) and Tobacco alternative products (TAL) were collected from January 3rd 2023 to December 31st,2023[3]. Work Life Balance (WLB) was collected from July 1st to December 31st, 2023

4.5 Content included in data files

The survey produces different data files:

- One-year data file
- Rapid response data file

5. SAMPLE DESIGN

5.1 Target population

In 2023, the CCHS covers the population 18 years of age and over living in the ten provinces and the three territories. The three territories are covered over two years of collection. Excluded from the survey's coverage are: persons living on reserves and other Aboriginal settlements in the provinces; full-time members of the Canadian Forces; the institutionalized population; and persons living in the Quebec health regions of Nunavik and Terres-Cries-de-la-Baie-James. Altogether, these exclusions represent less than 3% of the target population.

5.2 Health regions

For administrative purposes, each province is divided into health regions (HR) and each territory is designated as a single HR. Statistics Canada is sometimes asked to make minor changes to the boundaries of some of the HRs to correspond to the geography of the Census, or to better account for the health data needs determined by the new geographic boundaries. For CCHS 2023, data was collected in over 100 HRs in the ten provinces, as well as in one HR per territory (Appendix B).

5.3 Frames, household sampling strategies

In 2023, there were two major changes to the sampling of the CCHS. First, the youth population (12 to 17 years old) was not included in the sample. Second, a new targeted sample was introduced to meet the needs of the Data Disaggregation Action Plan (DDAP), the objective of which is to collect, analyze and disseminate more disaggregated data. Therefore, the 2023 CCHS sample consists of two frames: the area frame (which is the same as 2022) and the DDAP frame (which is new for 2023).

[3] The rapid response for SCH2 and TAL had a preliminary release using the data from the first two collection periods (January to July 2023) on February 29th, 2024.

5.3.1 Sampling of households from the area frame

The area frame used by the Labour Force Survey (LFS) is used as a sampling frame for the adult population. The LFS uses a two-stage sample design[4]. In the first stage, a sample of primary sampling units (PSUs), corresponding to geographical regions called clusters, is selected. In each selected PSU, a sample of dwellings is drawn at the second stage. The details about the area frame and the LFS sample design can be found here:
https://www150.statcan.gc.ca/n1/en/catalogue/71-526-x

Sampling of households from the area frame in the provinces

The provinces are divided into clusters, geographic regions generally formed of 100 to 600 dwellings. Each cluster is assigned to one of the six LFS rotation groups, and clusters are grouped into homogeneous, mutually exclusive, geographic or socio-economic groups called strata. The area frame contains all of the PSUs and their corresponding stratum.

In the first stage of sampling, a sample of clusters is drawn from each stratum. In the second stage, dwelling lists are prepared for each selected cluster, and dwellings are selected from these lists through systematic sampling.

Each cluster is divided into a set of starts, to allow for systematic sampling of dwellings within each cluster. For example, a cluster of 300 dwellings could be divided into 20 starts. On a list where the 300 dwellings are sorted according to the location of the dwellings, the first start would correspond to dwellings #1, #21, #41, …, #281, the second start would correspond to dwellings #2, #22, #42,…, #282, and so on. The sample size provided by a given start (15 dwellings in this case) is called the yield of the start.

The division of clusters into starts was done at the time of the latest LFS redesign in 2015, and the expected yield of each start was computed at that time. When a cluster is first selected by a survey, an updated list of the dwellings found in that cluster is defined. By applying the sampling rate that had been determined at the time of the LFS redesign to the latest count of dwellings, a more precise expected yield can then be obtained for the starts of the cluster.

Knowing the expected yield of each start, the objective of the CCHS sample reservation process is to determine how many starts have to be selected in each LFS stratum, in order for the expected yield to be large enough to obtain the allocated sample size for each of the CCHS strata. When selecting starts from a given LFS stratum, the same number of starts has to be selected within each of the six LFS rotation groups, meaning that the number of selected starts is always a multiple of six[5].

---

[4] Except for Prince Edward Island where a stratified Simple Random Sample design is used.

[5] A few LFS strata have two or three clusters for each rotation group, meaning that starts will come in groups of 12 or 18.

The LFS strata and the CCHS strata have different geographic boundaries, meaning that starts selected from a given LFS stratum can provide sample to different CCHS strata. Since we ensure that the total expected yield within each CCHS stratum will be at least the required sample size, and that starts are selected in groups of six, the total expected yield obtained from the selected starts will, aside from exceptions, typically exceed what is needed for each of the CCHS strata. When this is the case, a sub-sample of dwellings is selected within each CCHS stratum through a process called stabilization, in order to have a final sample size that corresponds to the target sample allocation for each CCHS stratum.

Sampling of households from the area frame in the three territories

For operational reasons, the LFS area frame sample design for the three territories was different. For each territory, the larger communities each have their own stratum while smaller communities are grouped into strata based on various characteristics (population, geographical information, proportion of Inuit and/or Aboriginal persons, and median household income). The LFS defined six design strata in the Yukon, ten in the Northwest Territories and ten in Nunavut. For strata consisting of a group of communities, the first stage of selection consisted of randomly selecting one community with a probability proportional to population size within each design stratum. Then, within the selected community, the second stage consisted of selecting households using the same sampling strategy as the one described above. If too many or too few dwellings were available for a community within a stratum, another community was selected for the CCHS. For larger communities with their own stratum, only one stage design was necessary where households were selected directly using the same sampling strategy described above.

It is worth mentioning that the frame for the CCHS covered 94% of the targeted population in the Yukon, 96% in the Northwest Territories and 93% in Nunavut.

5.3.2 Sampling of households from the 2021 Census of population for the DDAP initiative

For CCHS 2023, a new sample frame was added to meet the needs of the DDAP (Data Disaggregation Action Plan). The DDAP aims to support more representative data collection methods, enhance statistics on diverse populations to allow for intersectional analyses, and support government and societal efforts to address known inequalities by bringing considerations of fairness and inclusion into decision-making[6]. Disaggregated data based on the four employment equity groups should be provided for: Indigenous peoples, women, racialized populations and persons with disabilities. In 2021 and 2022 the sample size for CCHS was increased overall by approximately 50% to increase the number of DDAP group respondents.

For CCHS 2023, the DDAP initiative was enhanced by the availability of data from Census 2021. Census 2021 was chosen as an additional frame for the CCHS 2023 to allow stratification by population characteristics. Census 2021 data was used to target households with members who identified as belonging to groups that are part of the DDAP initiative. An extra sample of

[6] More information at https://www.statcan.gc.ca/en/trust/modernization/disaggregated-data

33,000 households was selected from Census 2021 long form data. These households were allocated across the last three collection periods of 2023. This extra sample was selected at the province and equity subgroup level.

5.4 Sample size and allocation

Although a primary objective of the CCHS is to produce reliable estimates for each HR given the allocated budget, the quality of the estimates for certain key characteristics at the provincial level is also deemed important. Therefore, the sample allocation strategy takes into account the needs at the provincial level. To limit response burden, a maximum sampling fraction of 10% was imposed to avoid sampling too many dwellings in smaller HRs also targeted by other surveys.

For the area frame sample in the ten provinces, a sample of about 72,000 respondents was allocated to the provinces, using a 0.75 power allocation based on the population size. Then, within each province, the sample was allocated to the HRs using a 0.35 power allocation based on the size of the population of the HR. In Ontario and Alberta, the HR sample sizes were allocated to the strata using an allocation proportional to the size of the strata.

For the area frame sample in the three territories, approximately 4,500 respondents are needed for a two-year cycle. The sample was allocated proportionally to the size of the strata.

The sample allocation strategy for the DDAP frame is significantly different. The four main DDAP groups (Indigenous, racialized populations, gender minorities and persons with disabilities) were divided into subgroups which vary by province. The annual DDAP sample was stratified by province and DDAP subgroup. The sample was allocated equally across the provinces and four main DDAP groups. Within the DDAP groups the sample was further allocated equally across the subgroups.

The number of respondents allocated to each stratum was then inflated to compensate for expected non-response and out-of-scope units. This inflated sample size is called the raw sample size and corresponds to the number of dwellings that need to be sent to collection to obtain the targeted number of respondents. The number of units sent to collection by province for the CCHS 2023 are available in Appendix C.

5.5 Sample allocation over the collection periods

In order to balance interviewer workload and to minimize possible seasonal effects on estimates, the initial sample size for each frame was allocated equally over all the four collection periods.

In the area frame (adults), each start selected within each HR was randomly assigned to a collection period accounting for a number of constraints related to field operations or weighting, while maintaining a uniform size for each period.

The DDAP sample was selected at the annual level. In order to optimize the collection activities, the annual sample was then assigned to collection periods to align with the geographic locations of the area frame sample clusters. This was done to minimize the travel of interviewers. DDAP sample units that did not geographically align with area fram clusters were allocated to a collection period randomly.

5.6 Sampling of interviewees

The selection strategy that was adopted accounted for user needs, cost, design efficiency, response burden and operational constraints. One person is selected per household using varying probabilities taking into account the age and the household composition. The selection probabilities resulted from simulations using various parameters in order to determine the optimal approach without causing extreme sampling weights.

Table 5.1 gives the person-level selection factors used to determine the probabilities of selection of individuals in sampled households by age group. For example, for a three-person household formed of two people between the ages of 35 and 64 and one 19-year-old, the 19-year-old would have a 1/2 chance of being selected (i.e., 20/(20+10+10)) while each of the 35 to 64 year-olds would have a 1/4 chance of being selected. To avoid extreme sampling weights, there is one exception to this rule: if the size of the household is greater than or equal to five then the person-level selection factor equals 1 for each individual in the household. Consequently, all people in that household have the same probability of being selected.

Table 5.1 Person-level selection factors, by age

Age	**18-34**	**35-49**	**50-64**	**65+**
Factor	20	10	10	20

6. DATA COLLECTION

6.1 Electronic questionnaires

In 2023, the CCHS used the Electronic Questionnaire (EQ) to collect data. This electronic questionnaire was securely accessed by respondents online or by Statistics Canada trained interviewers when a telephone or in person interview was conducted.

All selected units with a deliverable mailing address were sent an Introduction Letter (See subsection 6.4a for more details) with information on both how to access the Electronic Questionnaire for themselves or how to contact an interviewer to complete the response with interviewer assistance. A unique secure access code (SAC) was included in each letter to allow a secure method to access the survey online. For units without a mailable address, interviewers attempted either over the phone or in person interviews depending on what contact information was available for those units. Whether the survey was done online by the respondent themselves or by the interviewer, the content was the exact same.

The EQ was connected to a collection management system to track the status of a selected unit's response and the mode in which the survey was last accessed with or completed. This system also assists in managing calls and visits made by the interviewer to maximise interviewer efficiency.

Another advantage of the EQ was that it also allowed for custom interviews for every respondent based on their individual characteristics and survey responses. This includes:

- questions that are not applicable to the respondent are skipped automatically
- edits to check for inconsistent answers or out-of-range responses are applied automatically and on-screen prompts are shown when an invalid entry is recorded. Immediate feedback is given to the respondent and the interviewer is able to correct any inconsistencies.
- question text, including reference periods and pronouns, is customised automatically based on factors such as the age and sex of the respondent, the date of the interview and answers to previous questions.

6.2 CCHS application development

The electronic questionnaire consists of two parts, a principal component and a supplement component. The principal component determines household residency, demographics and respondent eligibility. Using this information it randomly selects the survey respondent and generate the supplement component which will be administered to the selected respondent. The supplement component of the EQ contains all health content of the survey which is made up of various health modules based around different subtopics. The electronic questionnaire collection application underwent three stages of testing: block, integrated and end to end.

Block level testing consists of independently testing each content module or "block" to ensure skip patterns, logic flows and text, in both official languages, are specified correctly.

Integrated testing occurs when all of the tested modules are added together, along with the Principal and Supplement components. This second stage of testing ensures that key demographic information is passed from the entry to the health content and exit components of the applications. It also ensures that variables affecting skip patterns and logic flows are correctly passed between modules within the health content.

End-to-end testing occurs when the fully integrated application is placed in a simulated collection environment. This stage of testing focuses on how the electronic questionnaire functions within the collection infrastructure at Statistics Canada. This allows for the testing of all technical aspects outside of the CCHS EQ application itself, such as the randomised selection of a respondent and the subsequent creation of the supplement questionnaire, testing the generation of invitation emails and the transmission and extraction of data. It also provided a final chance to detect errors within the electronic questionnaire itself.

Scenarios created by the Centre for Population health data (CPHD) are entered by regional offices, the Collection Planning and Research Division (CPRD) and other testers in CPHD. The returned data is then verified against the scripted cases for accuracy.

6.3 Interviewer training

Project managers, senior interviewers and interviewers from regional collection offices were sent self-study training packages before formal training sessions and the start of collection. The focus of the training sessions was to get interviewers comfortable using the CCHS 2023 application, and to familiarise interviewers with survey content and interviewing procedures specific to the CCHS.

The training focused on:

- goals and objectives of the survey
- survey methodology
- application functionality
- review of the questionnaire content and exercises with an emphasis on significant content changes
- interviewer techniques for maintaining response - complete exercises to minimise non-response
- use of mock interviews to simulate difficult situations and practise potential non-response situations
- survey management
- transmission procedures

One of the key aspects of the training was a focus on minimizing non-response. Exercises to minimise non-response were prepared for interviewers. The purpose of these exercises was to have the interviewers practice convincing reluctant respondents to participate in the survey. There was also a series of refusal avoidance workshops given to the senior interviewers responsible for refusal conversion in each regional collection office.

An offline version of the EQ application was also provided for interviewers to familiarize themselves with the application – in particular entry scripts and the content of the survey itself. Any functional gaps between practice applications and the actual application were covered by interviewer training and documentation.

6.4 The interview

Sample units were selected from a list frame, the Dwelling Universe File (DUF), for the population aged 18 and older.

In all selected dwellings from the DUF , a knowledgeable adult household member was asked to supply basic demographic information on all residents of the dwelling. One member of the

household was then selected for a more in-depth interview, which is referred to as the health content interview.

To ensure the quality of the data collected, interviewers were instructed to make every effort to conduct the interview with the selected respondent in privacy. In situations where this was unavoidable, the respondent was interviewed with another person present.

To ensure the best possible response rate attainable, many practices were used to minimise non-response, including:

a) Introductory letters
Before the start of each collection period, introductory letters and brochures explaining the purpose of the survey were sent to the sampled households. These explained the importance of the survey and provided examples of how CCHS data would be used.

b) Initiating contact
Interviewers were instructed to make all reasonable attempts to obtain interviews. Numerous call-backs were made at different times on different days. When the timing of the interviewer's call was inconvenient, an appointment was made to call back at a more convenient time.

c) Refusal conversion
For individuals who at first refused to participate in the survey, a letter was sent from the nearest Statistics Canada Regional Office to the respondent, stressing the importance of the survey and the household's collaboration. This was followed by a second call from a senior interviewer, a project supervisor or another interviewer to try to convince respondent of the importance of participating in the survey.

d) Language barriers
To remove language as a barrier to conducting interviews, each of the Statistics Canada Regional Offices recruited interviewers with a wide range of language competencies. When necessary, cases were transferred to an interviewer with the language competency needed to complete an interview.

e) Proxy interviews
In cases where the selected respondent was, for reasons of physical or mental health, incapable of completing an interview, another knowledgeable member of the household supplied information about the selected respondent. This is known as a proxy interview. While proxy interviewees were able to provide accurate answers to most of the survey questions, the more sensitive or personal questions were beyond the scope of knowledge of a proxy respondent. This resulted in some questions from the proxy interview being unanswered. Every effort was taken to keep proxy interviews to a minimum.

The Proxy interview (PR) module in the Supplement EQ prompts the interviewer to specifically identify whether the proxy interview is being conducted due to a physical or mental condition.

Interviewers are then asked to record the specific condition for either case. The variable ADM_PRX indicates whether a case was completed by proxy.

6.5 Field operations

The CCHS 2023 was designed to be four non-overlapping collection periods. Collection was scheduled to begin January 3rd, 2023, and ends on December 31st, 2023. Each cycle runs for approximately three months. However, the second cycle which was scheduled to start on April 3rd, 2023, and ends on June 30th, 2023, but this was extended to July 31st, 2023, for a number of reasons.

Regional collection offices were instructed to use the first few weeks of each collection period to resolve the majority of the sample, and use the remaining weeks to finalise the rest of the sample and to follow up on outstanding non-response cases. All cases were to have been attempted by the second week of each collection period. Sample files were sent approximately two weeks before the start of each collection period. A series of dummy cases were included with each sample. These cases were completed by senior interviewers for the purposes of ensuring that all data transmission procedures were working through the collection cycle.

For final response rates, refer to Appendix D.

6.6 Quality control and collection management

During the collection year, several methods are used to ensure data quality and to optimize collection. These included using internal measures to verify interviewer performance and the use of a series of ongoing reports to monitor various collection targets and data quality.

A system of validation was used whereby interviewers had their work validated on a regular basis by the senior collection staff. Each collection period, randomly selected cases were flagged in the sample. Validation procedures generally occurred during the first few weeks of a collection period to ensure that any issues were detected promptly. Interviewers were provided feedback by their supervisors on a regular basis.

7. DATA PROCESSING

7.1 Editing

Prior to 2022, most editing of the data was performed at the time of the interview by the computer-assisted interviewing (CAI) application. It was not possible for interviewers to enter out-of-range values and flow errors were controlled through programmed skip patterns. For example, CAI ensured that questions that did not apply to the respondent were not asked. In response to some types of inconsistent or unusual reporting, warning messages were invoked but no corrective action was taken at the time of the interview.

Due to the transition to the EQ collection mode, in-questionnaire edits have been reduced to a minimum to avoid disrupting the user experience. Where appropriate, edits are performed after data collection. Inconsistencies were usually reconciled by setting one or all variables in question to "not stated".

In EQ, respondents now have the choice to change any answers, pause the questionnaire, and to return to the questionnaire at any point until the survey deadline. Sometimes, a respondent does not return to a paused questionnaire, resulting in several unsubmitted questionnaires. All of these unsubmitted questionnaires are analyzed on a case-by-case basis according to pre-established completion criteria, and those that meet the criteria are deemed valid responses. The valid responses are then added to the final data set.

7.2 Flows

When data collected in the field is returned, any questions that were skipped will appear as blank on the preliminary data files. During processing, every blank variable is given one of two processing code values:

‘Valid Skip’ – VS
For variables of length 1, the value for VS will simply be 6. Otherwise, these values will appear on the file as a string of 9’s, followed by a 6, according to the length of the numerical variable or text field.

The code VS is applied whenever a question is skipped because it does not apply to the respondent (a population exclusion). The two main reasons for this code being used are optional module variables for respondents in provinces where that content was not selected; and questions that are skipped by a flow in the questionnaire since it did not apply, as with the Maternal Experiences questions for male respondents.

‘Not Stated’ – NS
For a variable of length 1, the value for NS will simply be 9. Otherwise, these will appear on the file as a string of 9’s, according to the length of the numerical variable or text field.

The code NS is applied whenever a question is skipped but could have been asked. These are respondents that are in the universe, but without an actual response. The two main reasons for this code being used are questions that are skipped because the interview is being completed by proxy; and questions that are skipped because a preceding question was itself ‘Not Stated’. For example, if a respondent refuses to answer CCC_Q05 (Do you have diabetes?), the questions that ask about the age of the diagnosis, insulin use, etc., are all set to ‘Not Stated’ since it is not known whether the respondent has diabetes.

7.3 Coding

Pre-coded answer categories were supplied for all suitable variables. Interviewers were trained to assign the respondent’s answers to the appropriate category.

If a respondent’s answer could not be easily assigned to an existing category, several questions also allowed the interviewer to enter a long-answer text in the “Other-specify” category. All such questions were closely examined during processing. For some of these questions, write-in responses were coded into one of the existing listed categories if the write-in information duplicated a listed category. In 2023, variables that were coded this way included RWC_05, GDRA_010, SOR_01, CAN_015, CAN_025, CAN_035, CCC_60 and PG_05. Other variables with other-specify responses for this cycle were not selected for coding. For all questions, the ‘Other-specify’ responses are taken into account when refining the answer categories for future cycles.

When write-in responses (“Other-specify”) are coded into an existing category during head office processing, it is possible that other questions in the questionnaire would become in scope to the respondent. Since those questions were never asked during the interview, the missing answers were set to ‘Not Stated’ in processing.

7.4 Creation of derived variables

To facilitate data analysis and to minimize the risk of error, a number of variables on the file have been derived using items found on the CCHS questionnaire. Derived variables generally have a "D", "G" or “F” in the fourth character of the variable name. In some cases, the derived variables are straightforward, involving the collapsing of response categories. In other cases, several variables have been combined to create a new variable. The *Derived Variables Documentation (DV)* provides details on how these more complex variables were derived. For more information on the naming convention, please go to Section 12.6.

7.5 Weighting

The principle behind estimation in a probability sample such as CCHS is that each person in the sample "represents", besides himself or herself, several other persons not in the sample. For example, in a simple random 2% sample of the population, each person in the sample represents 50 persons in the population. In the terminology used here, it can be said that each person has a weight of 50.

The weighting phase is a step that calculates, for each person, his or her associated sampling weight. This weight appears on the microdata files and must be used to derive meaningful estimates from the survey. For example, if the number of individuals who smoke daily is to be estimated, it is done by selecting the records referring to those individuals in the sample having that characteristic and summing the weights entered on those records. . Further details on the weighting process applied to the CCHS is described in section 8.

7.6 Income variables for 2023 - Linkage and Imputation

Since 2016, income variables in the 2023 CCHS master file are populated from one of three sources: tax records, respondent provided data, imputed data. Where respondents did not object to a link to their tax data and where a link to the appropriate tax records could be found, the income source variables (INC_005A-O, INC_Q025A-O), the main source of income (INC_015, INC_035), if they received a supplement for people with a disability (INC_010, INC_030) and the total income amounts (INC_021, INC_041) were determined from tax records. Where linkage to tax data was not feasible, respondent data was used. Note that in 2023 sources of income variables were not asked, the only income question asked was total household income. Also, as of the final collection period (October-December), no respondents were asked to report income . If neither linked tax data nor respondent reported data was available, an imputation of the total personal and total household income variables was conducted. The sources of incomes (INC_005A-O, INC_010, INC_015, INC_025A-O, INC_030, INC_035) were not imputed.

The variable INCFIMPP indicates the source used for the personal income variables and INCFIMPH indicates the source for household income variables. Starting in 2019, any tax source will be indicated as imputed. For more information concerning income on the master file, please refer to the document “CCHS_2023_Income_Master_File” available in the ‘Documentation’

folder of the 2023 CCHS. For more information concerning income on the share file, please refer to the document " CCHS_2023_Income_Share_File ".

7.7 Postal code imputation

A sampled unit's postal code is used together with the Postal Code Conversion File, to derive the rest of the geographical variables that are available on the CCHS data file. It is therefore important that all sampled units have a valid postal code. If a sampled unit's postal code is missing or invalid, it is usually imputed through a donor imputation process, although other imputation methods are sometimes used. The donor is chosen from the same geographical area, with as much precision as possible, as the unit with the missing or invalid postal code.

For the 2023 CCHS, approximately 5.9% of sampled units from the area frame had their postal codes imputed. For sampled units from the DDAP frame , approximately 3.1% had their postal codes imputed in 2023.

7.8 Gender coding

Gender refers to an individual's personal and social identity as a man, woman or non-binary person (a person who is not exclusively a man or a woman). Gender includes the following concepts:

- gender identity, which refers to the gender that a person feels internally and individually;
- gender expression, which refers to the way a person presents their gender, regardless of their gender identity, through body language, aesthetic choices or accessories (e.g., clothes, hairstyle and makeup), which may have traditionally been associated with a specific gender.

A person's gender may differ from their sex at birth, and from what is indicated on their current identification or legal documents such as their birth certificate, passport or driver's licence. A person's gender may change over time. Some people may not identify with a specific gender.

Given that the non-binary population is small, data aggregation to a two-category gender variable is sometimes necessary to protect the confidentiality of responses provided. In these cases, individuals in the category "non-binary persons", as well as those who do not respond to the question, are distributed into the other two gender categories and are denoted by the "+" symbol (e.g. men+, women+).

When releasing data that has been disaggregated by the recoded gender variable, users can include the notes:

- Men+: This category includes men (and/or boys), as well as some non-binary persons.
- Women+: This category includes women (and/or girls), as well as some non-binary persons.

In order to facilitate the use of the aggregated binary gender variable, the CCHS file includes a recoded variable: GENDER_2, already coded to the categories men+ and women+. Users are

advised to make use of this variable, rather than performing their own recoding, in order to be consistent with analysis and tables produced by Statistics Canada for the CCHS.

8. WEIGHTING

For estimates produced from survey data to be representative of the covered population, and not just the sample itself, users must incorporate the survey weights in their calculations. A survey weight is given to each respondent included in the final sample. This weight corresponds to the number of persons in the entire population that are represented by the respondent.

As described in section 5, the CCHS uses two sampling frames for its sample selection: an area frame for the Canadian population aged 18 and over and the 2021 Census for a targeted oversample of the Data Disaggregation Action Plan (DDAP) population aged 18 and over.

Weighting adjustments in 2023 are similar to those from 2022, where the area frame was also used.

The weighting strategy treats both the area frame and DDAP frame independently to come up with separate person-level weights for each of the frames used. The weights from the two frames are then integated into a single set of weights, jointly undergo a couple more adjustments (including person-level adjustments and being matched to known population totals) and become the final person-level weights.

8.1 Overview

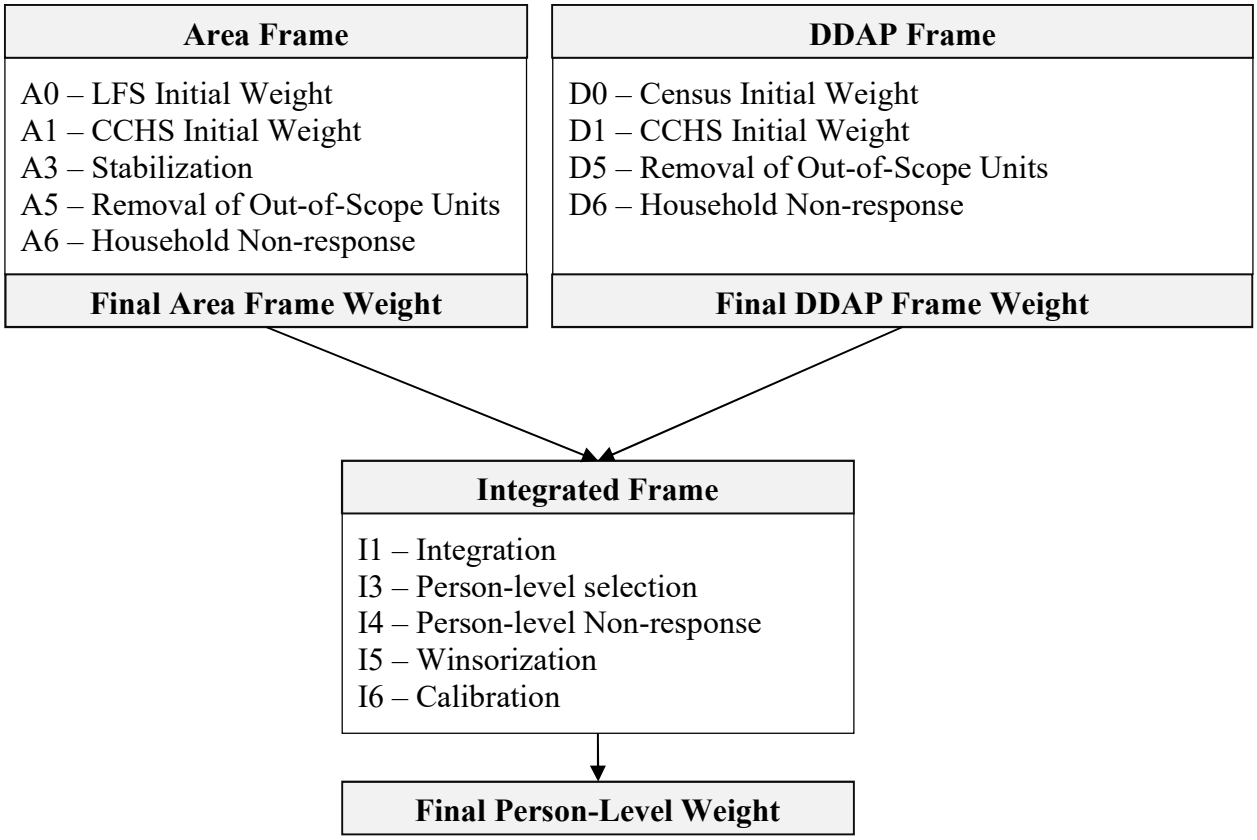
The following subsections describe the weighting process for the provinces. Units from both the area frame and the DDAP frame are treated separately up to and including the adjustment for person-level nonresponse. Sub-section 8.2 provides details on the weighting strategy for the area frame, while sub-section 8.3 deals with the strategy for the DDAP frame. The final weighting steps of that involve both frames, including person-level non-response, Winsorization and calibration, are discussed in sub-section 8.4.

Following the redesign of the CCHS in 2015, due to operational constraints, collection of a representative sample within some geographies are spread over a two-year period. For example, estimates in the three territories are typically representative over two years. As such, respondents in the territories are omitted from the one-year files and will only be included on the two-year files. There are some particularities associated with the weighting process for the territories, and this will be discussed in more detail in the documentation of the two-year CCHS files.

Many adjustments to unit weights are made throughout the weighting process. Adjustments applied to units from the area frame are enumerated from *A0* to *A*7, while adjustments applied to units from the DDAP frame are enumerated from D0 *to D7*. The final adjustments after

combining the adult and DDAP frames are called I1 to I6. Diagram A presents an overview of the most important of these adjustments that are part of the weighting strategy.

Diagram A - Weighting Strategy Overview



The subsections 8.2 to 8.4 present the details behind the steps shown in Diagram A.

8.2 Weighting of the area frame sample

A0 – LFS Initial Weight
The weighting process on the area frame sample begins with a weight provided by the Labour Force Survey (LFS). This weight is based on the LFS design since the CCHS area frame sample design is based on the LFS. The LFS design consists of a sample of dwellings within clusters selected from LFS strata, and the LFS initial weights reflect the probabilities of selection within LFS strata.

A1 – CCHS Initial Weight
In the initial adjustment A1, the LFS initial weight is adjusted to take into consideration the fact that the CCHS selects a sample of sufficient size for estimation at the Health Region level. To do so, the CCHS selects a different number of clusters than the LFS and can repeat the sampling of dwellings within the selected clusters. The resulting weight is called weight A1. For more details

about the selection mechanism, as well as a more complete definition of LFS strata and clusters, refer to Statistics Canada (2020)[7].

A3 – Stabilization

In some HRs, the increase in the sample size, as described in section 5, results in a larger sample than necessary. Stabilization is used to bring the sample size back down to the desired level. In order to minimize the dispersion of weights and thus reduce variability of estimates, stabilization is done in one of two ways within each HR. If the initial weights in each cluster within an HR are the same, the process consists of randomly sub-sampling dwellings at the HR level from the dwellings originally selected within each cluster. However, if the initial weights are not the same in each cluster within an HR, the process is done iteratively. First, the cluster with the lowest initial weight is selected and one unit from that cluster is removed. Then the weights are recalculated to adjust for the removed unit. Again the cluster with the lowest weight is selected and one unit is removed. This process is repeated until the desired sample size is achieved. An adjustment factor representing the effect of this stabilization is calculated in order to adjust the probability of selection appropriately. This factor, multiplied by weight in the previous step, produces weight A3.

A5 – Removal of Out-of-Scope Units

Among all dwellings sampled, a certain proportion are identified during collection as being out-of-scope. Dwellings that are demolished or under construction, vacant, seasonal or secondary, as well as institutions are examples of out-of-scope cases for the CCHS. Dwellings in which all residents are out-of-scope, for example, all are full-time members of the Armed Forces or visitors to Canada are also out-of-scope. All these out-of-scope dwellings and their associated weights are simply removed from the sample, as it is assumed that the weighted out-of-scope dwellings in the sample are representative of out-of-scope dwellings in the population. For dwellings whose scope status is unknown (not enough information was gathered to determine scope status), their weights are reduced by the out-of-scope rate of their CCHS stratum to account for some proportion of them being out-of-scope. This leaves a sample that is representative of in-scope dwellings or households. In-scope dwellings maintain the same weight as in the previous step, and is now called weight A5.

A6 – Household Nonresponse

During collection, a certain proportion of sampled households inevitably result in nonresponse. This usually occurs when a household refuses to participate in the survey, provides unusable data, or cannot be reached for an interview. Weights of the nonresponding households are redistributed to responding households within response homogeneity groups (RHGs), which are formed within province. In order to create the RHGs, a scoring method based on logistic regression models is used to determine the propensity to respond and these response probabilities are used to divide the sample into groups with similar response properties. The information

[7] Statistics Canada, 2020. *Guide to the Labour Force Survey*, Statistics Canada. Cat. No. 71-543-G.

available for non-respondents is limited so the regression models use characteristics such as the collection period, geographic information, language of questionnaire and variables from Statistics Canada's household survey frame. An adjustment factor is calculated within each RHGs as follows:

$$\frac{Sum\ of\ weights\ A5\ for\ all\ households}{Sum\ of\ weights\ A5\ for\ all\ responding\ households}$$

The weight from the previous step is multiplied by this adjustment factor to produce weight A6 for the responding households. Non-responding households are dropped from the process at this point.

8.3 Weighting of the DDAP frame sample

D0 – Initial Weight

The weighting process on the DDAP frame sample begins with the weight inherited from the 2021 long form Census. There are several weighting steps for the census, including a non-response adjustment and calibration. The CCHS uses the non-response adjusted weight as a starting weight, since calibration is performed later in the CCHS weighting process.

D1 – CCHS Initial Weight

The census non-response weight is adjusted based on the inverse of the dwelling selection probability in each DDAP stratum. That is, the number of units on the frame divided by the number of sampled dwellings in the year.

The initial weights (D0) are multiplied by this adjustment factor to produce weight D1.

D5 – Removal of Out-of-Scope Units

As in the area frame, among all dwellings sampled, a certain proportion are identified during collection as being out-of-scope. Examples include dwellings that are vacant, institutions, or dwellings in which all residents are out of scope. Similar to the methods used on the area frame, these cases are simply removed from the process, leaving in-scope dwellings in the sample. For dwellings with unknown scope, their weights are reduced by the out-of-scope rate in their stratum. In-scope dwellings maintain the same weight as the previous step, and are now called weight D5.

D6 - Household Non-response

The adjustment applied here to compensate for the effect of household non-response is identical to the one applied for the area frame units (adjustment A6). The adjustment factor calculated within each response homogeneity group is obtained as follows:

$$\frac{Sum\ of\ weights\ D5\ for\ all\ households}{Sum\ of\ weights\ D5\ for\ all\ responding\ households}$$

The weight of responding households is multiplied by this adjustment factor to produce the weight D6. Non-responding households are removed from the process at this point.

8.4 Final weight adjustments with area and DDAP frames combined

At this point in the process the sample from the two frames can be combined in anticipation of the final weight adjustments of person-level adjustments, Winsorization, and calibration. Since the population in the DDAP frame are part of the population of the area frame, it is necessary to perform an additional weight adjustment when combining the frames.

I1 – Integration of Frames

For 2023, the CCHS has introduced an additional sample from the DDAP frame that represent the same population that is also covered by the area frame. This step consists of combining the weights for the households common to both the DDAP frame and the area frame by applying a method of integration[8]. A provincial integration adjustment is calculated using four adjustment factors, calculated separately to represent each of the four main DDAP groups (α, β, γ and δ). Each adjustment factor can take on a value from 0 to 1. For each province, they are determined in such a way to represent the relative importance of each sample in the total group being integrated. The term "importance" here refers to the estimated effective sample size from each frame – the ratio of effective sample size (sample size divided by the design effect) in each of the two groups being integrated.

Additionally, each integration factor can be adjusted based on the probability that a household belongs to each DDAP group. When membership to a DDAP group is known, the probability of being in that DDAP group (or no DDAP group) can be set to 1, and the probability of being in the other groups (other DDAP groups or non-DDAP) is 0. For households with unknown domain membership, the probability of being in a DDAP group is estimated using the weighted proportion of responding households from the long-form census. Additionally, when a person in

[8] Skinner, C.J. and Rao, J.N.K. 1996. Estimation in Dual Frame Surveys with Complex Designs. Journal of the American Statistical Association. 91, 433, 349-356.

the dwelling has responded to the CCHS, their responses can be used to estimate the conditional probability of the household being in a DDAP group.

The associated probability, p, of being in each group (a=not any DDAP group and only area frame, d =disabled, r= racialized, g= gender minority and i=Indigenous) and the associated integration factors for each group ($\alpha = disability, \beta = racialized, \gamma = gender\ minority$ and $\delta = indigenous$) can be used to create the adjustment:

$$AdjI1 = p_{aj} + p_{dj}\alpha + p_{rj}\beta + p_{gj}\gamma + p_{ij}\delta$$

Note that for households with known DDAP group membership, ADJI1 simplifies to that group's integration factor, or 1 for households that are known to not be in any DDAP group. The previous weights from the area frame and DDAP frame are combined into one file at this point. The weight of the area frame units is multiplied by this factor AdjI1, while the weight of the DDAP frame units is multiplied by (1-AdjI1). This adjustment is called I1.

I3 – Person-level Selection

Since persons are the desired sampling units, the household-level weights computed to this point need to be converted to the person level. This weight is obtained by multiplying the household weight by the inverse of the probability of selection of the person selected in the household. This gives the weight I3. The probability of selection for an individual living in a dwelling selected from the area frame changes depending on the number of people in the household and their ages (see subsection 5.6 for more details).

I4 – Person-level Non-response

A CCHS questionnaire for the adult frame dwellings can be seen as a two-part process. First, a member of the household provides a complete roster of the people within the household. Second, the selected person completes the questionnaire. In some cases, only the first portion is completed, either because the selected person cannot be contacted, or because that selected person refuses to complete the electronic questionnaire or be interviewed. Such individuals are defined as person non-respondents and an adjustment factor must be applied to the weights of person respondents to account for this non-response. Using the same methodology that is used in the treatment of household non-response, the adjustment is applied within response homogeneity groups. In this process, the scoring method is used to define a response probability based on characteristics available for both respondents and non-respondents. In addition to characteristics previously used to model household-level non-response (geographic information and paradata), all characteristics collected when creating the roster of household members are available for the estimation of response probabilities. The probabilities are grouped into response homogeneity groups and the following adjustment factor is calculated within each group:

$$\frac{Sum\ of\ weights\ I3\ for\ all\ selected\ persons}{Sum\ of\ weights\ I3\ for\ all\ responding\ selected\ persons}$$

The weight for responding persons is multiplied by the above adjustment factor to produce weight I4. Non-responding persons are dropped from the weighting process from this point onward.

I5 – Winsorization

Following the series of adjustments applied to the respondents, some units may come out with extreme weights compared to other units of the same domain of interest. These units could have a large impact on the variance. In order to prevent this, the weight of these outlier units is adjusted downward using a "winsorization" trimming approach.

I6 – Calibration

The last step necessary to obtain the final CCHS weight is calibration (I6). Calibration is done using Statistics Canada's generalized estimation system, G-EST, to ensure that the sum of the final weights corresponds to the population estimates defined at the HR level, for all 10 age-sex groups of interest. The five age groups are 18-34, 35-49, 50-64, 65-74 and 75+, for both males and females. A minimum size of 15 respondents is required to calibrate at the HR by age and sex. When getting less than 15 respondents, some collapsing is done within the province and/or within sex. At the same time, weights are adjusted to ensure that each collection period is equally represented within the sample. Note that the calibration is done using the most up to date geography and may not match the geography used at the time of sampling.

The population estimates are projections based on the 2021 Census counts and counts of birth, death, immigration and emigration since that time. In a typical year, the average of these monthly estimates for each of the HR-age-sex post-strata by collection period is used to calibrate. Some changes to this calculation were made in 2023 due to changes in collection.

- The second and third collection periods overlapped in July for the adult population. To reflect this, census projections for this month were used in the estimation of the adult population size in both the first and second collection periods. The effect of this change on the estimated population size for the whole year was negligible.

The weight in the previous step is adjusted using G-EST to obtain the final weight I6. Weight I6 corresponds to the ***final CCHS person-level weight*** and can be found on the master file with the variable name WTS_M.

8.5 Creation of a share weight

Along with the master file which contain all CCHS respondents, a share file is created which contains only a portion of the original CCHS respondents. In 2023, with the introduction of the respondent electronic questionnaire in 2022, the share rate has decreased from above 90% to below 80%. The individuals on this share file have agreed to share their data with certain

partners. To compensate for the loss of some respondents from the file, the weights of the respondents who agreed to share must be adjusted by the factor:

$$\frac{Sum\ of\ weights\ I4\ for\ all\ respondents}{Sum\ of\ weights\ I4 for\ all\ respondents\ agreeing\ to\ share\ their\ data}$$

Similar to the nonresponse adjustments, this factor is calculated within response homogeneity groups, where in this case, individuals with similar estimated propensity to share will be grouped together. Weight I4 for respondents who agreed to share is multiplied by the above adjustment factor to produce a share weight. As with the Master weighting process, Winsorization and calibration (similar to adjustments I5 and I6) are applied to the share units. The final weight after these adjustments is called WTS_S.

9. DATA QUALITY

9.1 Response rates

In total, 127,322 of the selected units in the CCHS 2023 were in-scope for the survey[9], out of which a response was obtained for 51,640 individuals, resulting in a response rate of 40.6%. Appendix D provides the counts of in-scope units, the counts of respondents and the resulting response rates, by province. These figures are provided for the area frame and the DDAP frame separately.

9.2 Survey Errors

The estimates derived from this survey are based on a sample of individuals. Somewhat different figures might have been obtained if a complete census had been taken using the same questionnaire, interviewers, supervisors, processing methods, etc. than those actually used. The difference between the estimates obtained from the sample and the results from a complete count under similar conditions is called the *sampling error* of the estimate.

Errors which are not related to sampling may occur at almost every phase of a survey operation. Interviewers may misunderstand instructions, respondents may make errors in answering questions, the answers may be incorrectly entered on the computer and errors may be introduced in the processing and tabulation of the data. These are all examples of *non-sampling errors*.

[9] Among the units selected, some are not in-scope for the survey. They are, for example, dwellings selected from the list frame of dwellings but that turned out to be vacant, demolished or non-residential dwellingsThese units are identified during the data collection, otherwise they would have been excluded before the sample selection. These units are not considered in the calculation of response rates. This response rate excludes the three territories.

9.2.1 Non-Sampling Errors

Over a large number of observations, randomly occurring errors will have little effect on estimates derived from the survey. However, errors occurring systematically will contribute to biases in the survey estimates. Considerable time and effort was made to reduce non-sampling errors in the CCHS. Quality assurance measures were implemented at each step of data collection and processing to monitor the quality of the data. These measures included the use of highly skilled interviewers, extensive training with respect to the survey procedures and questionnaire, and the observation of interviewers to detect problems. Testing of the questionnaire was also an essential procedure to ensure that data collection errors were minimized.

A major source of non-sampling errors in surveys is the effect of *non-response* on the survey results. The extent of non-response varies from partial non-response (failure to answer just one or some questions) to total non-response. Partial non-response to the CCHS was minimal; once the questionnaire was started, it tended to be completed with very little non-response. Total non-response occurred either because a person refused to participate in the survey or because the interviewer was unable to contact the selected person. Total non-response was handled by adjusting the weight of persons who responded to the survey to compensate for those who did not respond. See section 8 for details on the weight adjustment for non-response.

9.2.2 Sampling Errors

Since it is an unavoidable fact that estimates from a sample survey are subject to sampling error, sound statistical practice calls for researchers to provide users with some indication of the magnitude of this sampling error. The basis for measuring the potential size of sampling errors is the standard deviation of the estimates derived from survey results. However, because of the large variety of estimates that can be produced from a survey, the standard deviation of an estimate is usually expressed relative to the estimate to which it pertains. This resulting measure, known as the coefficient of variation (CV) of an estimate, is obtained by dividing the standard deviation of the estimate by the estimate itself and is expressed as a percentage of the estimate. For example, suppose hypothetically that it is estimated that 25% of Canadians aged 12 and over are regular smokers and that this estimate is found to have a standard deviation of 0.003. Then the CV of the estimate is calculated as: (0.003/0.25) x 100% = 1.2%.

However, there has long been concerns regarding the suitability of using CVs as a measure of quality for proportions. For this type of estimates, confidence intervals should be used to get an indication of the sampling error. The standard deviation and a given probability threshold (usually 95%) make it possible to derive a confidence interval within which the true value of the estimate has a probability of being found. Confidence intervals have the advantage of being suitable for all types of estimates, including proportions, means, totals and differences. They are also easy to interpret.

For details on how to determine CVs and confidence intervals, see Section 11. For guidelines on dissemination, see sub-section 10.4.

10. GUIDELINES FOR TABULATION, ANALYSIS AND RELEASE

This section of the documentation offers guidelines to users for tabulating, analyzing, publishing or otherwise releasing any estimates derived from the survey files. With the aid of these guidelines, users of microdata should be able to produce figures that are in close agreement with those produced by Statistics Canada. They will also be able to develop currently unpublished figures in a manner consistent with these established guidelines. Methods to measure precision and quality are also described along with release guidelines to help decide when an estimate should be used in publication.

10.1 Rounding guidelines

In order that estimates for publication or other release derived from the data files (Master, Share or PUMF) correspond to those produced by Statistics Canada, users are urged to adhere to the following guidelines regarding the rounding of such estimates:

a) Estimates in the main body of a statistical table are to be rounded to the nearest hundred units using the normal rounding technique. In normal rounding, if the first or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is raised by one. For example, in normal rounding to the nearest 100, if the last two digits are between 00 and 49, they are changed to 00 and the preceding digit (the hundreds digit) is left unchanged. If the last digits are between 50 and 99 they are changed to 00 and the proceeding digit is incremented by 1;

b) Marginal sub-totals and totals in statistical tables are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units using normal rounding[1];

c) Averages, proportions, rates and percentages are to be computed from unrounded components (i.e., numerators and/or denominators) and then are to be rounded themselves to one decimal using normal rounding. In normal rounding to a single digit, if the final or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is increased by 1;

d) Sums and differences of aggregates (or ratios) are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units (or the nearest one decimal) using normal rounding;

e) In instances where, due to technical or other limitations, a rounding technique other than normal rounding is used resulting in estimates to be published or otherwise released that

differ from corresponding estimates published by Statistics Canada, users are urged to note the reason for such differences in the publication or release document(s);

f) Under no circumstances are unrounded estimates to be published or otherwise released by users. Unrounded estimates imply greater precision than actually exists.

10.2 Sample weighting guidelines for tabulation

The sample design used for this survey was not self-weighting. That is to say, the sampling weights are not identical for all individuals in the sample. When producing simple estimates, including the production of ordinary statistical tables, users must apply the proper sampling weight. If proper weights are not used, the estimates derived from the data file cannot be considered to be representative of the survey population and will not correspond to those produced by Statistics Canada.

Users should also note that some software packages might not allow the generation of estimates that exactly match those available from Statistics Canada, because of their treatment of the weight field. If options are available, users should ensure that they specify that the weight is a sample weight rather than a frequency weight.

10.2.1 Definitions: categorical estimates, quantitative estimates

Before discussing how the survey data can be tabulated and analyzed, it is useful to describe the two main types of point estimates of population characteristics that can be generated from the data files.

Categorical estimates:
Categorical estimates are estimates of the number or percentage of the surveyed population possessing certain characteristics or falling into some defined category. The number of individuals who smoke daily is an example of such an estimate. An estimate of the number of persons possessing a certain characteristic may also be referred to as an estimate of an aggregate.

Example of categorical question:

In the past 30 days, how often did … smoke cigarettes? ***(CSS_020)***

_ *Every day*
_ *Less than once a day, but at least once a week*
_ *Less than once a week, but at least once in the past month*
_ *Not at all*

Quantitative estimates:
Quantitative estimates are estimates of totals or of means, medians and other measures of central tendency of quantities based upon some or all of the members of the surveyed population.

An example of a quantitative estimate is the average number of cigarettes smoked per day by individuals who smoke daily. The numerator is an estimate of the total number of cigarettes smoked per day by individuals who smoke daily, and its denominator is an estimate of the number of individuals who smoke daily.

Example of quantitative question:

*How many cigarettes does … usually smoke each day? (**CSS_025**)*
|_|_|_| *Number of cigarettes*

10.2.2 Tabulation of categorical estimates

Estimates of the number of people with a certain characteristic can be obtained from the data file by summing the final weights of all records possessing the characteristic of interest.

Proportions and ratios of the form $\hat{X} / \hat{Y}$ are obtained by:

a) summing the final weights of records having the characteristic of interest for the numerator ($\hat{X}$);
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then
c) dividing the numerator estimate by the denominator estimate.

10.2.3 Tabulation of quantitative estimates

Estimates of sums or averages for quantitative variables can be obtained using the following three steps (only step *a* is necessary to obtain the estimate of a sum):

a) multiplying the value of the variable of interest by the final weight and summing this quantity over all records of interest to obtain the numerator ($\hat{X}$);
b) summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$); then
c) dividing the numerator estimate by the denominator estimate.

For example, to obtain the estimate of the average number of cigarettes smoked each day by individuals who smoke daily, first compute the numerator ($\hat{X}$) by summing the product between the value of variable **CSS_025** and the weight **WTS_M.** Next, sum this value over those records with a value of "every day" to the variable **CSS_020**. The denominator ($\hat{Y}$) is obtained by summing the final weight of those records with a value of "every day" to the variable **CSS_020**. Divide ($\hat{X}$) by ($\hat{Y}$) to obtain the average number of cigarettes smoked each day by daily smokers.

10.3 Guidelines for statistical analysis

The CCHS is based upon a complex design, with stratification and multiple stages of selection, and unequal probabilities of selection of respondents. Using data from such complex surveys presents problems to analysts because the survey design and the selection probabilities affect the

estimation and variance calculation procedures that should be used. The use of the survey weights is the first step to ensuring that the proper results are obtained.

While many analysis procedures found in statistical packages allow weights to be used, the meaning or definition of the weight in these procedures can differ from what is appropriate in a sample survey framework. The end result is that, while in many cases the estimates produced by the packages are correct, the variances and the statistics based on those variances are almost meaningless. To get the proper results, the weights must be correctly interpreted by the software package and the proper variance estimation techniques must be used.

To ensure that results from this complex design are meaningful, there are several options. The first and most appropriate option is to use the variance calculation options outlined in Chapter 11 based on the bootstrap methodology. Second, survey specific procedures exist in many packages that properly interpret the weight. For example, in SAS, PROC SURVEYMEANS would provide results that are more appropriate as the bootstrap weights are used for the variance estimates than the results produced from PROC MEANS. PROC MEANS can use the survey weights to correctly adjust estimates, taking into account the unequal weighting design, but it tends to underestimate the variances in not using the bootstrap weights and assuming the design is simple random sampling. Please consult the bootstrap variance estimation document found in the CCHS release package for more information. It is recommended that data users use software packages and functions that can incorporate both sample weights and bootstrap weights while performing their analyses.

Only the options outlined in Chapter 11 based on the bootstrap methodology will take the stratification, clustering and multiple frame design into account when calculating the variance.

10.4 Release guidelines

Since the estimates obtained from the survey are based on a sample, there is variability in the values obtained in the sense that a different sample could result in different results. To take this into consideration, users should first ensure that there are enough observations to properly estimate the statistic and also to estimate the variance. Once the variance is obtained, users should ensure that the variance is reasonable enough that the estimate can properly be interpreted as being near the true population value.

Before releasing and/or publishing any estimates, analysts should consider the quality level of the estimate. While data quality is affected by both sampling and non-sampling errors, this section covers quality in terms of sampling error. It is considered a best practice at Statistics Canada to report the sampling error of an estimate through its 95% confidence interval (CI). The confidence interval should be released with the estimate, in the same table as the estimate. In addition to the confidence intervals, estimates are categorized into one of three release categories:

<u>Category A</u>

The estimate and confidence interval can be released with no warning. Data users should use the 95% confidence interval to assess whether the quality of the estimate is sufficient. Note that the 'A' is not a quality indicator; it should not be released with the estimate.

Category E

The estimate and confidence interval should be flagged with the letter E (or some similar identifier) and accompanied by a quality warning to use the estimate with caution. Data users should use the 95% confidence interval to assess whether the quality of the estimate is sufficient.

Category F

The estimate and confidence interval are not recommended for release. They are deemed of such poor quality, that they are not fit for any use; they contain a very high level of instability, making them unreliable and potentially misleading. If analysts insist on releasing estimates of poor quality, even after being advised of their accuracy, the estimates should be accompanied by a disclaimer. Analysts should acknowledge the warnings given and undertake not to disseminate, present or report the estimates, directly or indirectly, without this disclaimer. The estimates should be flagged with the letter F (or some similar identifier) and the following warning should accompany the estimates and confidence intervals: "Please be warned that these estimates and confidence intervals [flagged with the letter F] do not meet Statistics Canada's quality standards. Conclusions based on these data will be unreliable, and may be invalid."

The rules for assigning an estimate to a release category depends on the type of estimate. Additionally, note that required sample sizes according to the release rules are subject to change year over year.

10.4.1 Master or share file

For users of the master or share files, it is recommended to have at least 10 observations with the characteristic of interest and 20 in the domain if a proportion is being calculated. With enough observations, the user can proceed to calculating the variance and the coefficients of variation using the bootstrap weights provided with the data along with the appropriate software to do the analysis. The variances can be used to calculate the CVs and the confidence intervals. Confidence intervals aid in assessing the reliability of an estimate in regard to the sampling variability. The CV provides a relative measure of the sampling error as a proportion of the estimate. Estimates should be vetted using the guidelines in Section 10.4. CVs will increase as the variability of an estimate increases, and decrease as an estimate is more precise. However, an estimate can be precise while still having a large CV, specifically if the estimate has a value close to 0. Examining the confidence interval of the estimate will provide further indication of the quality of the estimate in terms of the variability. Long confidence intervals indicate less precision in the estimate while smaller confidence intervals indicate greater precision. When assessing the trustworthiness of sample proportions, the confidence intervals of estimates should be taken into account. It is considered best practices at Statistics Canada to report the quality of an estimate with its 95% confidence interval.

10.4.2 Release rules for estimated proportions and estimated counts

Estimated proportions are computed from binary variables. Estimated proportions are estimates of the proportion of persons/households with a characteristic of interest (e.g., estimated proportion of daily smokers in the general population). Estimated proportions can also be computed from categorical variables: that is, estimates of the proportion of persons/household who belong to a category.

The release rules for estimated proportions are based on sample size and the effective sample size as estimated by the variable of interest. Table 10.1 provides the release rules for the CCHS. Estimates of 0% or 100% should not be published.

It is important to note that the effective sample size $\frac{(1-\hat{p})}{(\hat{p})CV^2}$, can equivalently be expressed using the estimated variance: $\frac{\hat{p}(1-\hat{p})}{\widehat{Var(\hat{p})}}$

Table 10.1: General rules for proportions

Sample Size (n)	Release Category	Action
$n_2 \geq 100$ AND $\frac{(1-\hat{p})}{(\hat{p})CV^2} \geq 60$	A*	Release with no warning; users should use CI as quality indicator
Otherwise	E	Release with quality warning; users should use CI as quality indicator
$n_2 < 50$ OR $\frac{(1-\hat{p})}{(\hat{p})CV^2} < 30$ OR **$n_1 < 10$**	F	Suppress the estimate and its CI for quality reasons

* Note that 'A' is not a quality indicator; it should not be released with the estimate. The 95% confidence interval is the quality indicator.

For estimated proportions, n_1 is defined as the unweighted count of the number of respondents in the numerator and n_2 is defined as the unweighted count of the number of respondents in the denominator of the proportion.

10.4.3 Release rules for estimated counts, estimated means and estimated totals

Estimated counts are estimates of the total number of persons/households with a characteristic of interest; in other words, they are the weighted sum of a binary variable (e.g., estimated number of smokers). The release rules for the estimated counts, means and totals of quantitative variables or amounts are based on the sample size and on the CV of the estimate, shown in Table 10.2.

Table 10.2: General rules for counts, means and totals

Sample Size (n)	Release Category	Action
$n \geq 100$ AND $CV \leq 25\%$	A*	Release with no warning; users should use CI as quality indicator
$50 \leq n < 100$ AND $CV \leq 35\%$	E	Release with quality warning; users should use CI as quality indicator
Otherwise	F	Suppress the estimate and its CI for quality reasons

* Note that 'A' is not a quality indicator; it should not be released with the estimate. The 95% confidence interval is the quality indicator.

For estimated counts, n is defined as the unweighted count of the number of respondents with nonzero values that contribute to the estimate. For estimated means, n is defined as the unweighted count of the number of respondents that contribute to the estimate including values of zero. For estimated totals, n is defined as the unweighted count of the number respondents with nonzero values that contribute to the estimate.

10.4.4 Release rules for differences and ratios

In order to assign a release category for an estimated difference or estimated ratio between two estimates, the analyst must first determine the release category of each of the two estimates using the rules described above. Next, the release category of the estimated difference or the estimate of change is assigned the lower release category of the two estimates; this can be specified as follows:

- If one or both estimates are category F estimates, then assign the estimated difference to category F and suppress it.
- Otherwise, if one or both estimates are category E estimates, then assign the estimated difference to category E.
- If both estimates are category A estimates, then assign the estimated difference to category A.

10.4.5 Additional Rules Regarding Confidence intervals

The above release rules should suppress most estimates and confidence intervals of poor quality. There are also two conditions that indicate that a confidence interval is of poor quality. An estimate and its confidence interval should be assigned to release category F if either of the following two conditions are true:

- The lower bound of the 95% confidence interval is equal to the upper bound of the interval; in other words, the confidence interval is of length zero. (Exceptions are if the estimate corresponds to a calibration control total.)
- The lower bound or upper bound of the 95% confidence interval is not a plausible value for the estimate. For example, the lower bound for an estimated proportion is negative.

11. APPROXIMATE SAMPLING VARIABILITY TABLES

For a quick approximation of coefficients of variation that will be applicable to a wide variety of categorical estimates produced from the Share file, a set of Approximate Sampling Variability Tables will be produced. These "look-up" tables allow the user to obtain an approximate coefficient of variation based on the size of the estimate calculated from the survey data. From

this coefficient of variation, the confidence interval will be computed. Table 11.1 shows the CCHS 2017 look-up table at the Canada level as an example.

Table 11.1, CCHS Approximate Sampling Variability Tables, 2017 Share

NUMERATOR OF PERCENTAGE ('000)	ESTIMATED PERCENTAGE 0.1%	1.0%	2.0%	5.0%	10.0%	15.0%	20.0%	25.0%	30.0%	35.0%	40.0%	50.0%	70.0%	90.0%
1	126.8	126.2	125.5	123.6	120.3	116.9	113.4	109.8	106.1	102.2	98.2	89.7	69.5	40.1
2	89.6	89.2	88.8	87.4	85.1	82.7	80.2	77.7	75.0	72.3	69.5	63.4	49.1	28.4
3	73.2	72.9	72.5	71.4	69.5	67.5	65.5	63.4	61.3	59.0	56.7	51.8	40.1	23.2
4	63.4	63.1	62.8	61.8	60.2	58.5	56.7	54.9	53.1	51.1	49.1	44.8	34.7	20.1
5	56.7	56.4	56.1	55.3	53.8	52.3	50.7	49.1	47.5	45.7	43.9	40.1	31.1	17.9
6	51.7	51.5	51.3	50.5	49.1	47.7	46.3	44.8	43.3	41.7	40.1	36.6	28.4	16.4
7	47.9	47.7	47.5	46.7	45.5	44.2	42.9	41.5	40.1	38.6	37.1	33.9	26.3	15.2
8	44.8	44.6	44.4	43.7	42.5	41.3	40.1	38.8	37.5	36.1	34.7	31.7	24.6	14.2
9	42.3	42.1	41.8	41.2	40.1	39.0	37.8	36.6	35.4	34.1	32.7	29.9	23.2	13.4
10	40.1	39.9	39.7	39.1	38.0	37.0	35.9	34.7	33.6	32.3	31.1	28.4	22.0	12.7
11	38.2	38.0	37.9	37.3	36.3	35.3	34.2	33.1	32.0	30.8	29.6	27.0	20.9	12.1
12	36.6	36.4	36.2	35.7	34.7	33.8	32.7	31.7	30.6	29.5	28.4	25.9	20.1	11.6
13	35.2	35.0	34.8	34.3	33.4	32.4	31.5	30.5	29.4	28.4	27.2	24.9	19.3	11.1
14	33.9	33.7	33.6	33.0	32.2	31.2	30.3	29.4	28.4	27.3	26.3	24.0	18.6	10.7
15	32.7	32.6	32.4	31.9	31.1	30.2	29.3	28.4	27.4	26.4	25.4	23.2	17.9	10.4
16	31.7	31.5	31.4	30.9	30.1	29.2	28.4	27.5	26.5	25.6	24.6	22.4	17.4	10.0
17	30.7	30.6	30.4	30.0	29.2	28.4	27.5	26.6	25.7	24.8	23.8	21.7	16.8	9.7
18	29.9	29.7	29.6	29.1	28.4	27.6	26.7	25.9	25.0	24.1	23.2	21.1	16.4	9.5
19	29.1	28.9	28.8	28.4	27.6	26.8	26.0	25.2	24.3	23.5	22.5	20.6	15.9	9.2
20	28.3	28.2	28.1	27.6	26.9	26.1	25.4	24.6	23.7	22.9	22.0	20.1	15.5	9.0
21	27.7	27.5	27.4	27.0	26.3	25.5	24.8	24.0	23.2	22.3	21.4	19.6	15.2	8.8
22	27.0	26.9	26.8	26.4	25.7	24.9	24.2	23.4	22.6	21.8	20.9	19.1	14.8	8.6
23	26.4	26.3	26.2	25.8	25.1	24.4	23.7	22.9	22.1	21.3	20.5	18.7	14.5	8.4
24	25.9	25.8	25.6	25.2	24.6	23.9	23.2	22.4	21.7	20.9	20.1	18.3	14.2	8.2
25	25.4	25.2	25.1	24.7	24.1	23.4	22.7	22.0	21.2	20.4	19.6	17.9	13.9	8.0
30	23.1	23.0	22.9	22.6	22.0	21.3	20.7	20.1	19.4	18.7	17.9	16.4	12.7	7.3
35	21.4	21.3	21.2	20.9	20.3	19.8	19.2	18.6	17.9	17.3	16.6	15.2	11.7	6.8
40	20.0	20.0	19.9	19.5	19.0	18.5	17.9	17.4	16.8	16.2	15.5	14.2	11.0	6.3
45	18.9	18.8	18.7	18.4	17.9	17.4	16.9	16.4	15.8	15.2	14.6	13.4	10.4	6.0
50	17.9	17.8	17.8	17.5	17.0	16.5	16.0	15.5	15.0	14.5	13.9	12.7	9.8	5.7
55	17.1	17.0	16.9	16.7	16.2	15.8	15.3	14.8	14.3	13.8	13.2	12.1	9.4	5.4
60	16.4	16.3	16.2	16.0	15.5	15.1	14.6	14.2	13.7	13.2	12.7	11.6	9.0	5.2
65	15.7	15.7	15.6	15.3	14.9	14.5	14.1	13.6	13.2	12.7	12.2	11.1	8.6	5.0
70	15.2	15.1	15.0	14.8	14.4	14.0	13.6	13.1	12.7	12.2	11.7	10.7	8.3	4.8
75	14.6	14.6	14.5	14.3	13.9	13.5	13.1	12.7	12.3	11.8	11.3	10.4	8.0	4.6
80	14.2	14.1	14.0	13.8	13.5	13.1	12.7	12.3	11.9	11.4	11.0	10.0	7.8	4.5
85	13.7	13.7	13.6	13.4	13.0	12.7	12.3	11.9	11.5	11.1	10.7	9.7	7.5	4.3
90	13.4	13.3	13.2	13.0	12.7	12.3	12.0	11.6	11.2	10.8	10.4	9.5	7.3	4.2
95	13.0	12.9	12.9	12.7	12.3	12.0	11.6	11.3	10.9	10.5	10.1	9.2	7.1	4.1
100	12.7	12.6	12.6	12.4	12.0	11.7	11.3	11.0	10.6	10.2	9.8	9.0	6.9	4.0
125	11.3	11.3	11.2	11.1	10.8	10.5	10.1	9.8	9.5	9.1	8.8	8.0	6.2	3.6
150	10.3	10.3	10.3	10.1	9.8	9.5	9.3	9.0	8.7	8.3	8.0	7.3	5.7	3.3
200	9.0	8.9	8.9	8.7	8.5	8.3	8.0	7.8	7.5	7.2	6.9	6.3	4.9	2.8
250	8.0	8.0	7.9	7.8	7.6	7.4	7.2	6.9	6.7	6.5	6.2	5.7	4.4	2.5
300	7.3	7.3	7.2	7.1	6.9	6.8	6.5	6.3	6.1	5.9	5.7	5.2	4.0	2.3
350	********	6.7	6.7	6.6	6.4	6.2	6.1	5.9	5.7	5.5	5.3	4.8	3.7	2.1
400	********	6.3	6.3	6.2	6.0	5.8	5.7	5.5	5.3	5.1	4.9	4.5	3.5	2.0
450	********	5.9	5.9	5.8	5.7	5.5	5.3	5.2	5.0	4.8	4.6	4.2	3.3	1.9
500	********	5.6	5.6	5.5	5.4	5.2	5.1	4.9	4.7	4.6	4.4	4.0	3.1	1.8
750	**********	4.6	4.5	4.4	4.3	4.1	4.0	3.9	3.7	3.6	3.3	2.5	1.5
1000	**********	4.0	3.9	3.8	3.7	3.6	3.5	3.4	3.2	3.1	2.8	2.2	1.3
1500	**********	3.2	3.2	3.1	3.0	2.9	2.8	2.7	2.6	2.5	2.3	1.8	1.0
2000	**********	2.8	2.7	2.6	2.5	2.5	2.4	2.3	2.2	2.0	1.6	0.9
3000	**********	2.3	2.2	2.1	2.1	2.0	1.9	1.9	1.8	1.6	1.3	0.7
4000	**********	1.9	1.8	1.8	1.7	1.7	1.6	1.6	1.4	1.1	0.6
5000	**********	1.7	1.6	1.6	1.5	1.4	1.4	1.3	1.0	0.6
6000	**********	1.5	1.5	1.4	1.4	1.3	1.3	1.2	0.9	0.5
7000	**********	1.4	1.3	1.3	1.2	1.2	1.1	0.8	0.5
8000	**********	1.2	1.2	1.1	1.1	1.0	0.8	0.4
9000	**********	1.2	1.1	1.1	1.0	0.9	0.7	0.4
10000	**********	1.1	1.0	1.0	0.9	0.7	0.4
12500	**********	0.9	0.8	0.6	0.4
15000	**********	0.8	0.7	0.6	0.3

The coefficients of variation (CV) are derived using the variance formula for simple random sampling and incorporating a factor which reflects the multi-stage, clustered nature of the sample design. This factor, known as the ***design effect,*** was determined by first calculating design effects for a wide range of characteristics and then choosing, for each table produced, a conservative value among all design effects relative to that table. The value chosen was then used to generate a table that applies to the entire set of characteristics.

The Approximate Sampling Variability Tables, along with the design effects, the sample sizes and the population counts that were used to produce them, are provided in the document *Approximate Sampling Variability Tables*, which is available to the share file users. All coefficients of variation in the Approximate Sampling Variability Tables are approximate and, therefore, unofficial. Options concerning the computation of exact coefficients of variation are discussed in sub-section 11.6. Analysis using bootstrap weights is the method that should be employed for more precise results.

Remember: As indicated in the "Release guidelines" in Section 10.4, if the number of observations on which an estimate is based is less than 10, the weighted estimate should not be released regardless of the value of the coefficient of variation. Coefficients of variation based on small sample sizes are too unpredictable to be adequately represented. Using the tables to estimate coefficients of variation are even less precise than using the bootstrap weights, so in this case, a minimum of 30 observations is required.

11.1 How to use the CV tables to obtain confidence limits

Although coefficients of variation are widely used, a more intuitively meaningful measure of sampling error is the confidence interval of an estimate. It is considered best practices at Statistics Canada to report the quality of an estimate with its 95% confidence interval. A confidence interval constitutes a statement on the level of confidence that the true value for the population lies within a specified range of values. For example a 95% confidence interval can be described as follows: if sampling of the population is repeated indefinitely, each sample leading to a new confidence interval for an estimate, then in 95% of the samples the interval will cover the true population value.

Using the standard error of an estimate, confidence intervals for estimates may be obtained under the assumption that under repeated sampling of the population, the various estimates obtained for a population characteristic are normally distributed about the true population value. Under this assumption, the chances are about 68 out of 100 that the difference between a sample estimate and the true population value would be less than one standard error, about 95 out of 100 that the difference would be less than two standard errors, and about 99 out of 100 that the differences would be less than three standard errors. These different degrees of confidence are referred to as the confidence levels.

Confidence intervals for an estimate, $\hat{X}$, are generally expressed as two numbers, one below the estimate and one above the estimate, as $\left(\hat{X} - k, \hat{X} + k\right)$, where k is determined depending upon the level of confidence desired and the sampling error of the estimate.

Confidence intervals for an estimate can be calculated directly from the Approximate Sampling Variability Tables by first determining from the appropriate table the coefficient of variation of the estimate $\hat{X}$, and then using the following formula to convert to a confidence interval CI:

$$CI_X = [\ \hat{X} - z\ \hat{X}\ \alpha_{\hat{X}}, \hat{X} + z\ \hat{X}\ \alpha_{\hat{X}}\ ]$$

Where $\alpha_{\hat{X}}$ is determined coefficient of variation for $\hat{X}$, and

z = 1 if a 68% confidence interval is desired
z = 1.6 if a 90% confidence interval is desired
z = 2 if a 95% confidence interval is desired
z = 3 if a 99% confidence interval is desired.

Note: "Release guidelines" presented in section 10.4 which apply to the estimate also apply to the confidence interval. For example, if the estimate is not releasable, then the confidence interval is not releasable either.

11.2 How to use the CV tables for categorical estimates

The following rules should enable the user to determine the approximate coefficients of variation from the Sampling Variability Tables for estimates of the number, proportion or percentage of the surveyed population possessing a certain characteristic and for ratios and differences between such estimates.

Rule 1: Estimates of numbers possessing a characteristic (aggregates)
The coefficient of variation depends only on the size of the estimate itself. On the appropriate Approximate Coefficients of Variations Table, locate the estimated number in the left-most column of the table (headed "Numerator of Percentage") and follow the asterisks (if any) across to the first figure encountered. Since not all the possible values for the estimate are available, the smallest value which is the closest must be taken (as an example, if the estimate is equal to 1,700 and the two closest available values are 1,000 and 2,000, the first has to be chosen). This figure is the approximate coefficient of variation.

Rule 2: Estimates of proportions or percentages of people possessing a characteristic
The coefficient of variation of an estimated proportion (or percentage) depends on both the size of the proportion and the size of the numerator upon which the proportion is based. Estimated proportions are relatively more reliable than the corresponding estimates of the numerator of the proportion when the proportion is based upon a sub-group of the population. This is due to the fact that the coefficients of variation of the latter type of estimates are based on the largest entry in a row of a particular table, whereas the coefficients of variation of the former type of estimators are based on some entry (not necessarily the largest) in that same row. (Note that in the tables the CVs decline in value reading across a row from left to right). For example, the estimated <u>proportion</u> of individuals who smoke daily out of those who smoke at all is more reliable than the estimated <u>number</u> who smoke daily.

When the proportion (or percentage) is based upon the total population covered by each specific table, the CV of the proportion is the same as the CV of the numerator of the proportion. In this case, this is equivalent to applying Rule 1.

When the proportion (or percentage) is based upon a subset of the total population (e.g., those who smoke at all), reference should be made to the proportion (across the top of the table) and to the numerator of the proportion (down the left side of the table). Since not all the possible values for the proportion are available, the smallest value which is the closest must be taken (for example, if the proportion is 23% and the two closest values available in the column are 20% and 25%, 20% must be chosen). The intersection of the appropriate row and column gives the coefficient of variation.

Rule 3: Estimates of differences between aggregates or percentages

The standard error of a difference between two estimates is approximately equal to the square root of the sum of squares of each standard error considered separately. That is, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is:

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$. This formula is accurate for the difference between independent populations or subgroups, but is only approximate otherwise. It will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 4: Estimates of ratios

In the case where the numerator is a subset of the denominator, the ratio should be converted to a percentage and Rule 2 applied. This would apply, for example, to the case where the denominator is the number of individuals who smoke at all and the numerator is the number of individuals who smoke daily out of those who smoke at all.

Consider the case where the numerator is not a subset of the denominator, as for example, the ratio of the number of individuals who smoke daily or occasionally as compared to the number of individuals who do not smoke at all. The standard deviation of the ratio of the estimates is approximately equal to the square root of the sum of squares of each coefficient of variation considered separately multiplied by $\hat{R}$, where $\hat{R}$ is the ratio of the estimates ($\hat{R} = \hat{X}_1 / \hat{X}_2$). That is, the standard error of a ratio is:

$$\sigma_{\hat{R}} = \hat{R} \sqrt{\alpha_1^2 + \alpha_2^2}$$

Where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively.

The coefficient of variation of $\hat{R}$ is given by $\sigma_{\hat{R}} / \hat{R} = \sqrt{\alpha_1^2 + \alpha_2^2}$. The formula will tend to overstate the error, if $\hat{X}_1$ and $\hat{X}_2$ are positively correlated and understate the error if $\hat{X}_1$ and $\hat{X}_2$ are negatively correlated.

Rule 5: Estimates of differences of ratios

In this case, Rules 3 and 4 are combined. The CVs for the two ratios are first determined using Rule 4, and then the CV of their difference is found using Rule 3.

11.3 Examples of using the CV tables for categorical estimates to calculate confidence intervals

The following "real life" examples are included to assist users in applying the foregoing rules. Unrounded numbers are used in the examples to facilitate the step by step instructions. These examples use the 2017 CCHS CV tables (Table 11.1). Users should still follow the rounding guidelines outlined in section 10.1 before publishing numbers.

Example 1: Estimates of numbers possessing a characteristic (aggregates)

Suppose that a user estimates that 3,665,449 individuals smoke daily in Canada and that the corresponding unweighted count is 6,789. How does the user determine the 95% confidence interval of this estimate?

1) Refer to the CANADA level CV table.
2) The estimated aggregate (3,665,449) does not appear in the left-hand column (the "Numerator of Percentage" column), so it is necessary to use the smallest figure closest to it, namely 3,000,000.
3) The coefficient of variation for an estimated aggregate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.
4) Following the steps in 11.1, we have that:
   $\hat{X}$ = 3,665,449
   Z = 2
   $\alpha_{\hat{X}}$ = 0.023 is the coefficient of variation of this estimate as determined from the tables.
   CI_X = {3,665,449 - (2) (3,665,449) (0.023), 3,665,449 + (2) (3,665,449) (0.023)}
   CI_X = {3,496,838, 3,834,060}

5) Because the unweighted count of 6,789 is higher than 100, and the CV of 2.3% is less than 25%, according to table 10.2, this estimate can be released after applying the rounding rules with no warning; users should use CI as quality indicators.

Example 2: Estimates of proportions or percentages possessing a characteristic

Suppose that the user estimates that 3,665,449 / 5,151,237 = 71.2% of individuals in Canada who smoke at all smoke daily. And that the sampling size of the denominator is 9,234. How does the user determine the 95% confidence interval of this estimate?

1) Refer to the CANADA level CV table.
2) Because the estimate is a percentage which is based on a subset of the total population (i.e., individuals who smoke at all, that is to say, daily or occasionally), it is necessary to use both the percentage (71.2%) and the numerator portion of the percentage (3,665,449) in determining the coefficient of variation.
3) The numerator (3,665,449) does not appear in the lefthand column (the "Numerator of Percentage" column) so it is necessary to use the smallest figure closest to it, namely 3,000,000. Similarly, the percentage estimate does not appear as any of the column headings, so it is necessary to use the smallest figure closest to it, 70.0%.
4) The figure at the intersection of the row and column used, namely 1.3% is the coefficient of variation (expressed as a percentage) to be used.
5) Following the steps in 11.1, we have that:
   $\hat{X}$ = 71.2%
   Z = 2
   $\alpha_{\hat{X}}$ = 0.013 is the coefficient of variation of this estimate as determined from the tables.
   CI_X = {0.712 - (2) (0.712) (0.013), 0.712 + (2) (0.712) (0.023)}
   CI_X = {0.693, 0.731}

6) Because the sample size of the denominator 9,234 is higher than 100 and $\frac{(1-\hat{p})}{(\hat{p})CV^2} = \frac{1-0.712}{(0.712)(0.013^2)} = 2{,}393.46$ is higher than 60, according to table 10.1, this estimate can be released with no warning; users should use CI as quality indicators.

Example 3: Estimates of differences between aggregates or percentages

Suppose that a user estimates that, among men, 2,067,955 / 15,115,880 = 13.7% smoke daily (estimate 1), while for women, this percentage is estimated at 1,597,494 / 15,528,483 = 10.3% (estimate 2). For estimate 1, the sample size of the denominator is 23,546 and for the estimate 2, the sample size of the denominator is 27,834. How does the user determine the confidence interval of the difference between these two estimates?

1) Using the CANADA level CV table in the same manner as described in example 2 gives the CV for estimate 1 as 2.7% (expressed as a percentage), and the CV for estimate 2 as 3.1% (expressed as a percentage).

2) Using rule 3, the standard error of a difference ($\hat{d} = \hat{X}_2 - \hat{X}_1$) is :

$$\sigma_{\hat{d}} = \sqrt{(\hat{X}_1 \alpha_1)^2 + (\hat{X}_2 \alpha_2)^2}$$

where $\hat{X}_1$ is estimate 1, $\hat{X}_2$ is estimate 2, and α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The standard error of the difference $\hat{d}$ = (0.137 - 0.103) = 0.034 is:

$$\sigma_{\hat{d}} = \sqrt{[(0.137)(0.027)]^2 + [(0.103)(0.031)]^2}$$

= 0.00489

3) The coefficient of variation of $\hat{d}$ is given by $\sigma_{\hat{d}} / \hat{d}$ = 0.00489 / 0.034 = 0.144.
4) So the approximate coefficient of variation of the difference between the estimates is 14.4% (expressed as a percentage).
5) Following the steps in 11.1, we have that:
   $\hat{X}$ = 0.034
   Z = 2
   $\alpha_{\hat{X}}$ = 0.144 is the coefficient of variation of this estimate as determined from the tables.
   CI_X = {0.034 - (2) (0.034) (0.144), 0.034+ (2) (0.034) (0.144)}
   CI_X = {0.0242, 0.0438}

6) Alternatively, we could have computed the confidence interval by using the standard error of the difference:
   CI_X = {0.034 - (2) (0.00489), 0.034+ (2) (0.00489)}

7) According to the "Release guidelines" presented in Section 10.4, this estimate can be published with no warning. Both sample sizes of the denominator are greater than 100 (23,546 and 27,834), and both $\frac{(1-\hat{p})}{(\hat{p})CV^2}$ are greater than 60 (estimate 1: $\frac{(1-\hat{p})}{(\hat{p})CV^2} = \frac{1-0.137}{(0.137)(0.027^2)} = 8{,}640.97$ and estimate 2: $\frac{(1-\hat{p})}{(\hat{p})CV^2} = \frac{1-0.103}{(0.0.103)(0.031^2)} = 9{,}062.16$). Both estimates can be published with no warning, as given by table 10.1. Given the release rules for differences outlined in section 10.4.4, the difference of the estimates can be published with no warning.

Example 4: Estimates of ratios

Suppose that the user estimates that 3,665,449 individuals smoke daily, and the corresponding unweighted count is 6,789, while 1,485,788 individuals smoke occasionally, and the corresponding unweighted count is 2,144. The user is interested in comparing the estimate of daily to occasional smokers in the form of a ratio. How does the user determine the coefficient of variation of this estimate?

1) First of all, this estimate is a ratio estimate, where the numerator of the estimate (= $\hat{X}_1$) is the number of individuals who smoke occasionally. The denominator of the estimate (= $\hat{X}_2$) is the number of individuals who smoke daily.
2) Refer to the CANADA level CV table.

3) The numerator of this ratio estimate is 1,485,788. The smallest figure closest to it is 1,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 4.0%.
4) The denominator of this ratio estimate is 3,665,449. The figure closest to it is 3,000,000. The coefficient of variation for this estimate (expressed as a percentage) is found by referring to the first non-asterisk entry on that row, namely, 2.3%.
5) So the approximate coefficient of variation of the ratio estimate is given by rule 4:

$$\alpha_{\hat{R}} = \sqrt{\alpha_1^2 + \alpha_2^2},$$

That is,

$$\alpha_{\hat{R}} = \sqrt{(0.040)^2 + (0.023)^2}$$

= 0.046

where α_1 and α_2 are the coefficients of variation of $\hat{X}_1$ and $\hat{X}_2$ respectively. The obtained ratio of occasional to daily smokers is 1,485,788 / 3,665,449 which is 0.41:1. The coefficient of variation of this estimate is 4.6% (expressed as a percentage). The 95% confidence interval is (37.2%, 44.8%)

6) The estimate is releasable with no qualifications, according to the "Release guidelines" presented in Section 10.4 because for each estimate the corresponding unweighted count is bigger than 100, and each estimate has a CV less than 25%. As indicated in table 10.2, both estimates can be published with no warning and, given the release rules for ratios outlined in section 10.4.4, the ratio of the estimates can be published with no warning.

11.4 How to use the CV tables to do a Z-test

Standard errors may also be used to perform hypothesis testing, a procedure for distinguishing between population parameters using sample estimates. The sample estimates can be numbers, averages, percentages, ratios, etc. Tests may be performed at various levels of significance, where a level of significance is the probability of concluding that the characteristics are different when, in fact, they are identical.

Let $\hat{X}_1$ and $\hat{X}_2$ be sample estimates for two characteristics of interest. Let the standard error on the difference $\hat{X}_1 - \hat{X}_2$ be $\sigma_{\hat{d}}$. If the ratio of $\hat{X}_1 - \hat{X}_2$ over $\sigma_{\hat{d}}$ is between -2 and 2, then no conclusion about the difference between the characteristics is justified at the 5% level of significance. If however, this ratio is smaller than -2 or larger than +2, the observed difference is significant at the 5% level.

11.5 Example of using the CV tables to do a Z-test

Let us suppose we wish to test, at 5% level of significance, the hypothesis that there is no difference between the proportion of men who smoke daily AND the proportion of women who

smoke daily. From example 3, sub-section 11.2, the standard error of the difference between these two estimates was found to be = 0.00489. Hence,

$$z = \frac{\hat{X}_1 - \hat{X}_2}{\sigma_{\hat{d}}} = \frac{0.137 - 0.103}{0.0048995} = \frac{0.034}{0.0048995} = 6.87$$

Since Z = 6.95 is greater than 2, it must be concluded that there is a significant difference between the two estimates at the 5% level of significance. Note that the two sub-groups compared are considered as being independent, so the test is valid.

11.6 Exact variances or coefficients of variation

All coefficients of variation in the Approximate Sampling Variability Tables (CV Tables) are indeed approximate and, therefore, unofficial.

The computation of exact coefficients of variation is not a straightforward task since there is no simple mathematical formula that would account for all CCHS sampling frame and weighting aspects. Therefore, other methods such as resampling methods must be used in order to estimate measures of precision. Among these methods, the bootstrap method is the one recommended for analysis of CCHS data.

Many statistical packages allow for analyses using the bootstrap weights. More details are provided in section 10.3 and in the document on bootstrap variance estimation.

There are a number of reasons why a user may require an exact variance. A few are given below. Firstly, if a user desires estimates at a geographic level other than those available in the tables (for example, at the rural/urban level), then the CV tables provided are not adequate. Coefficients of variation of these estimates may be obtained using "domain" estimation techniques through the exact variance program.

Secondly, should a user require more sophisticated analyses such as estimates of parameters from linear regressions or logistic regressions, the CV tables will not provide correct associated coefficients of variation. Although some standard statistical packages allow sampling weights to be incorporated in the analyses, the variances that are produced often do not take into account the stratified and clustered nature of the design properly, whereas the exact variance program would do so.

Thirdly, for estimates of quantitative variables, separate tables are required to determine their sampling error. Since most of the variables for the CCHS are primarily categorical in nature, this has not been done. Thus, users wishing to obtain coefficients of variation and confidence intervals for quantitative variables can do so through the exact variance program.

Coefficients of variation produced by the tables are based on a wide range of variables and are therefore considered crude, whereas the exact variance program would give an exact coefficient

of variation associated with the variable in question. Confidence intervals computed using coefficients of variation produced by the tables are therefore not considered to be precise.

12. MICRODATA FILES: DESCRIPTION, ACCESS AND USE

The CCHS produces three types of microdata files: master files, share files and public use microdata files (PUMF). Table 12.1 includes the list of all available 2023 collection period data files.

12.1 Master files

The master files contain all variables and all records from the survey for an associated time frame. These files are accessible at Statistics Canada for internal use and in Statistics Canada's Research Data Centres (RDC) and are also available for custom tabulation requests.

12.1.1 Research Data Centre

The RDC Program enables researchers to use the survey data in the master files in a secure environment in several universities across Canada. Researchers must submit research proposals that, once approved, give them access to the RDC. For more information, please consult the following web page: https://www.statcan.gc.ca/eng/rdc/index

12.1.2 Custom tabulations

Another form of access to data is by custom tabulation services offered by the Client Services of the Centre for Population Health Data. This service is offered on a cost recovery basis. It allows users who do not possess knowledge of tabulation software products to get custom results based on the master file data. The results are screened for confidentiality and reliability concerns before release. For more information, please contact Client Services at 613-951-1746 or by e mail at statcan.hd-ds.statcan@canada.ca.

12.1.3 Remote access

The remote access service to the survey master files is another way to have access to these data if, for some reason, the user cannot access a Research Data Centre but have an approved research proposal. A researcher can be supplied with a synthetic or 'dummy' master file and a corresponding record layout. With these tools, the researcher can develop his or her own set of analytical computer programs. The code for the custom tabulations is then sent via e-mail to statcan.hd-ds.statcan@statcan.gc.ca. The code will then be transferred into Statistics Canada's internal secured network and processed using the appropriate master file of CCHS data. Estimates generated will be released to the user, subject to meeting the guidelines for analysis and release outlined in Section 10 of this document. Results are screened for confidentiality and reliability concerns and then the output is returned to the client. There is no charge for this service.

12.1.4 Real Time Remote Access (RTRA)

Users can access CCHS data using the Real Time Remote Access (RTRA) system, enabling fast access to Statistics Canada microdata. The RTRA system is an on-line remote access facility allowing users to run auto-coded SAS programs (created through a graphical interface), in real-time, against micro-data sets located in a central and secure location. Researchers using the RTRA system do not gain direct access to the micro-data and cannot view the content of the micro-data file. Instead, users submit the generated SAS programs to extract results in the form of frequency tables. As RTRA researchers cannot view the micro-data, becoming a deemed employee of Statistics Canada is no longer necessary (as is the case for RDC access). This relationship is the basis that allows the RTRA to service its clients rapidly. Please see https://www.statcan.gc.ca/eng/rtra/rtra for more information.

12.2 Share files

The share files contain all variables and all records of CCHS respondents who agreed to share their data with Statistic Canada's partners, which are Health Canada and the Public Health Agency of Canada, as well as the most provincial and territorial ministries of health and l'Institut de la statistique du Québec (ISQ). The share file is released only to these organizations. Personal identifiers are removed from the share files to respect respondent confidentiality and only provided to the provincial and territorial partners. Users of these share files must first certify that they will not disclose, at any time, any information that might identify a survey respondent.

12.3 Public use microdata files

The public use microdata files (PUMF) are developed from the master files using a technique that balances the need to ensure respondent confidentiality with the need to produce the most useful data possible at the health region level. The PUMF must meet stringent security and confidentiality standards required by the *Statistics Act* before they are released for public access. To ensure that these standards have been achieved, each PUMF goes through a formal review and approval process by an executive committee of Statistics Canada.

Variables most likely to lead to identification of an individual are deleted from the data file or are collapsed to broader categories. Due to the risk of disclosure household weights are not included on the PUMF.

There is no charge to access the PUMF in a post-secondary educational institution that is part of the Data Liberation Initiative, a partnership between post-secondary institutions and Statistics Canada for improving access to Canadian data resources. For more information on the Data Liberation Initiative please see http://www.statcan.gc.ca/eng/dli/dli.
The CCHS PUMF files are also free of charge to any researcher that contacts Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@statcan.gc.ca.

Table 12.1 2022 CCHS data files

Reference Period	Files	File Type	File name	Sampling weight	Bootstrap weights file	Variables included	Records included
2023	Main file - provinces	Master	HS.txt	WTS_M	bsw.txt	All core and all theme modules for the provinces	All respondent records in the ten provinces
		Share		WTS_S			Records of all respondents who agreed to share their data
		Share income	HS_inc.txt	N/A	N/A	Income content for share partners, kept on separate file	Income data for all respondents who agreed to share their data
January to December 2023	Rapid response file – (SCH2/TAL) [10]	Master	HS_SCH2TAL.txt	WTS_M	Bsw.txt	All core modules and questions in the SCH2 & TAL rapid response	All respondent records in the three territories
		Share		WTS_S			Records of all respondents who agreed to share their data
		Share income	HS_SCH2TAL_inc.txt	N/A	N/A	Income content for share partners, kept on separate file	Income data for all respondents who agreed to share their data
July to December 2022	Rapid response file (WLB)	Master	HS_WLB.txt	WTS_M	bsw.txt	All core modules and questions in the WLB rapid response	All respondent records in the ten provinces
		Share		WTS_S			Records of all respondents who agreed to share their data
		Share income	HS_WLBINC.txt	N/A	N/A	Income content for share partners, kept on separate file	Income data for all respondents who agreed to share their data

12.4 How to use the CCHS data files: annual data file or two-year data file?

Users who have access to share files or master files normally have the choice of using one-year or two-year data files, depending on availability. Decisions about which period to use in a given data analysis should be guided by the level of detail and the quality required. With a one-year file, estimates will not always be available because of the quality associated with limited sample sizes.

Before interpreting and using a CCHS estimate, it is recommended to make sure that the estimates meets the rules for minimum sample sizes outlined in Chapter 10.

[10] Note that a preliminary version of this rapid response was released using the data from just the first two collection periods, covering January to June 2023.

This will not be possible for all rare characteristics and detailed domains with one-year files. Instead, users will have to rely on available two-year files or pool data with other cycles of CCHS (as appropriate depending on the design) in order to obtain multi-year files. For more information on combining CCHS cycles into a multi-year file, please see https://www150.statcan.gc.ca/n1/pub/82-003-x/2009001/article/10795-eng.pdf

Where the use of either a one-year or two-year file is viable, the user should consider the trade-off between accuracy and timeliness. If it is important to reflect the current characteristics of a population as closely as possible, the one-year file would be preferable. However, with the increased sample size, more detailed estimates and analyses can be carried out with a two-year file.

Due to significant changes between 2021 and 2022 to the survey content and methodology, as well as because of the increased risk of bias that will result from the vastly different response rates (2021 had a response rate of 24%, while 2022 has a rate of 43%), users are advised to take caution when combining data from 2021 with 2022. Users who choose to pool with 2021 data, should assess the methodological and content differences and determine the appropriateness for pooling as it pertains to their analysis. The 2023 cycle of CCHS (which has virtually the same design as 2022) will be released in mid-2024, and users will have the opportunity at that point to pool data to increase sample size for analysis with a consistently designed and collected survey.

12.5 Use of weight variable

The weight variable **WTS_M** represents the sampling weight for key survey files. For a given respondent, the sampling weight can be interpreted as the number of people the respondent represents in the Canadian population. This weight must always be used when computing statistical estimates in order to make inference at the population level possible. The production of unweighted estimates is not recommended. The sample allocation, as well as the survey design specifics can cause such results to not correctly represent the population. Refer to section 8 on weighting for a more detailed explanation on the creation of this weight. The weight variable **WTS_M** must be used for regional analyses.

The Food security (FSC) module, included in certain reference period data files, measures concepts that apply not only to the respondent's situation, but also to that of the respondent's entire household. The Home Care Services (HMC) module also has some content related to the household level. Depending on the level of analysis, the analysis of the variables may require use of a weight calculated to represent the number of Canadian households, rather than the number of persons. This weight variable **WTS_MHH** is found in a separate file (HS_HHWT.txt). It can be used in place of the variable **WTS_M** for household analyses at the national and provincial levels (for the share file it is WTS_SHH which replaces WTS_S for household level analysis).

12.6 Variable naming convention

The variable naming convention adopted allows data users to easily use and identify the data based on the module and variable type. The CCHS variable naming convention fulfils two

requirements: to restrict variable names to a maximum of eight characters for ease of use by analytical software products and to identify easily conceptually identical variables from one survey collection period to the next. Questions to which changes are made between two collection periods, and where the changes alter the concept measured by the question, are entirely renamed to avoid any confusion in the analysis.

A variable (REFPER, format = YYYYMM-YYYYMM) was added to the microdata files in order to identify the beginning and the end of the reference during which data included in the file were collected. This variable will be useful, notably for users wanting to use data from several collection periods at a time. Therefore, variable names for identical modules or questions from one collection year to the next (e.g., 2015 and 2017) will be the same. However, some minor changes have been made to variable names since the 2015 redesign.

The naming convention used for variables in the 2023 CCHS use up to eight characters. The variable names are generally structured as follows:

Positions 1 to 3 :	Module/questionnaire section name
Position 4 :	Variable type (underscore, C,D, F or G)
Positions 5 to 8 :	Question number and answer option for multiple response questions in increments of five (e.g. 005, 010, 015)

An example that shows the structure of the variable name for question 005, Current smoking status, is CSS_005 :

Positions 1 to 3 :	CSS	Current smoking status module
Position 4 :	_	(underscore = collected data)
Position 5 to 8 :	005	Question number

Positions 1 to 3 generally contain the acronyms for each of the modules. These acronyms appear beside the module names given in the table in Appendix A. It should be noted that some module acronyms consist of four characters. Variables names from these modules are still bound by a maximum of 8 characters.
Position 4 designates the variable type based on whether it is a variable collected directly from a questionnaire question ("_"), derived ("D"), flag ("F"), or grouped ("G") variable.
In general, the last four positions (5 to 8) follow the variable numbering used on the questionnaire.

The letter "Q" used to represent the word "question" is removed, and all question numbers are presented in a two or three digit format. For example, question Q005A in the questionnaire becomes simply 005A, and question Q010 becomes simply 010.

Table 12.2 Designation of codes used in the 4th position of the CCHS variable names

_	Collected variable	A variable that appears directly on the questionnaire
C	Coded variable	A variable coded from one or more collected variables (e.g., SIC, Standard Industrial Classification code)
D	Derived variable	A variable calculated from one or more collected or coded variables, usually calculated during head office processing (e.g., Health Utility Index)
F	Flag variable	A variable calculated from one or more collected variables (like a derived variable), but usually calculated by the data collection computer application for later use during the interview (e.g., work flag)
G	Grouped variable	Collected, coded, suppressed or derived variables collapsed into groups (e.g., age groups). Mostly used for the PUMF.

For questions that have more than one response option, the final position in the variable naming sequence is represented by a letter. For this type of question, new variables were created to differentiate between a "yes" or "no" answer for each response option. For example, if Q010 had 4 response options, the new questions would be named Q010A for option 1, Q010B for option 2, Q010C for option 3, and Q010D for option 4. If only options 2 and 3 were selected, then Q010A = No, Q010B = Yes, Q010C = Yes and Q010D = No.

12.7 Data dictionaries

Separate data dictionary reports, including variable names, concepts, universe statements and frequencies, are provided for the main master and share files and each of the rapid response files (if applicable). All variables on these files will appear in the data dictionary report, though some may not include a frequency (SampleID for example).

When a variable includes the frequencies for responses, the report will show four columns:

1. Content – The text label for each of the response categories.
2. Code – The numerical value associated with each category.
3. Sample – The unweighted frequency on the file of respondents with each response.
4. Population – The weighted frequency for the response.

For each module in the data dictionary reports, a flag is used to indicate the availability of the module for each respondent. When the flag is equal to 2 (No), all variables in the module have “not applicable” values. For example, the DOCCC variable indicates if the Chronic conditions module applies to a given respondent.

APPENDIX A - CANADIAN COMMUNITY HEALTH SURVEY (CCHS) - CONTENT OVERVIEW – 2023

Appendix A - Canadian Community Health Survey (CCHS)
Content availability by province and territory - 2023

		2023												
Acronym	**Module Name**	**NL (10)**	**PEI (11)**	**NS (12)**	**NB (13)**	**QC (24)**	**ON (35)**	**MB (46)**	**SK (47)**	**AB (48)**	**BC (59)**	**YK[1] (60)**	**NWT[1] (61)**	**NU[1] (62)**
ABM	Indigenous identity	C	C	C	C	C	C	C	C	C	C	C	C	C
ADL	Activities of daily living	O		O		O			O				O	
ALC	Alcohol use	C	C	C	C	C	C	C	C	C	C	C	C	C
ALW	Alcohol use during the past week	T	T	T	T	T	T	T	T	T	T	T	T	T
CAN	Cannabis use	C	C	C	C	C	C	C	C	C	C	C	C	C
CCC	Chronic conditions	C	C	C	C	C	C	C	C	C	C	C	C	C
CC1	Chronic conditions 1			O		O	O	O	O			O	O	
CMH	Consultations about mental health			O	O	O		O	O	O		O	O	
COVA[2]	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
COVB[2]	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
COVC	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
COVD	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
COVE	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
COVF	COVID-19	T	T	T	T	T	T	T	T	T	T	T	T	T
CSS	Current smoking status	C	C	C	C	C	C	C	C	C	C	C	C	C
DEP	Mental health	O	O	O	O	O	O	O	O	O	O	O		O
EAH	Eating habits											T	T	T
ECV	Electronic cigarettes and vaping	C	C	C	C	C	C	C	C	C	C	C	C	C
FLU	Flu shots	T	T	T	T	T	T	T	T	T	T	T	T	T
FSC	Food security	C	C	C	C	C	C	C	C	C	C	C	C	C
GDRA	Sex and gender	C	C	C	C	C	C	C	C	C	C	C	C	C
GEN	General health	C	C	C	C	C	C	C	C	C	C	C	C	C
HMC	Home care services	C	C	C	C	C	C	C	C	C	C	C	C	C

HWT	Height and weight	C	C	C	C	C	C	C	C	C	C	C	C	C
HUI	Health utility index	T	T	T	T	T	T	T	T	T	T	T	T	T
IM	Immigration and citizenship	C	C	C	C	C	C	C	C	C	C	C	C	C
INC	Total household income	C	C	C	C	C	C	C	C	C	C	C	C	C
ADMC	Administration information	C	C	C	C	C	C	C	C	C	C	C	C	C
HN	Health number (sub-block of ADMC)	C	C	C	C	C	C	C	C	C	C	C	C	C
LAN	Language	C	C	C	C	C	C	C	C	C	C	C	C	C
LBF	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LM	Telework	T	T	T	T	T	T	T	T	T	T	T	T	T
LMAM	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA3	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA4	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA5	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LMA6	Labour market activities	C	C	C	C	C	C	C	C	C	C	C	C	C
LSM	Life satisfaction	C	C	C	C	C	C	C	C	C	C	C	C	C
LTC	Long-term condition	C	C	C	C	C	C	C	C	C	C	C	C	C
MA	Main activity	C	C	C	C	C	C	C	C	C	C	C	C	C
EDC	Main activity	C	C	C	C	C	C	C	C	C	C	C	C	C
MEX	Maternal experiences	O	O	O		O		O	O	O	O	O	O	
MPA	Moderate physical activity	T	T	T	T	T	T	T	T	T	T	T	T	T
OPA	Other physical activity	T	T	T	T	T	T	T	T	T	T	T	T	T
PG	Population group	C	C	C	C	C	C	C	C	C	C	C	C	C
PMS	Positive mental health scale		O		O						O			O
PRS	Pregnancy	C	C	C	C	C	C	C	C	C	C	C	C	C
RHC	Regular health care provider	C	C	C	C	C	C	C	C	C	C	C	C	C
SCH2	Smoking - stage of change 2	RR	RR	RR	RR	RR	RR	RR	RR	RR	RR			
SOR	Sexual orientation	C	C	C	C	C	C	C	C	C	C	C	C	C
SPS	Social provisions	O	O	O	O	O	O		O	O	O		O	
SPU	Smoking - past use	C	C	C	C	C	C	C	C	C	C	C	C	C
TAL	Tobacco alternative products	RR	RR	RR	RR	RR	RR	RR	RR	RR	RR			
UPE	Use of protective equipment			O		O			O	O		O		

WLB	Work family balance	RR	RR	RR	RR	RR	RR	RR	RR	RR	RR			

Legend - Content Type	
C	Core
T	Theme
O	Optional
RR	Rapid response
	Not available

APPENDIX B - AVAILABLE GEOGRAPHY IN THE MASTER AND SHARE FILES AND THEIR CORRESPONDING
CODES: CANADA, PROVINCES, TERRITORIES, HEALTH REGIONS AND PEER GROUPS

Appendix B - Available geography in the master and share files and their corresponding codes: Canada, provinces, territories, health regions and peer groups

0	Canada
10	Newfoundland and Labrador
1020-G	Eastern Urban Zone
1021-B	Eastern Rural Zone
1022-B	Central Zone
1023-B	Western Zone
1024-E	Labrador-Grenfell Zone
11	Prince Edward Island
1100-I	Prince Edward Island
12	Nova Scotia
1201-B	Zone 1 - Western
1202-B	Zone 2 - Northern
1203-B	Zone 3 - Eastern
1204-G	Zone 4 - Central
13	New Brunswick
1301-I	Zone 1 (Moncton area)
1302-I	Zone 2 (Saint John area)
1303-I	Zone 3 (Fredericton area)
1304-B	Zone 4 (Edmundston area)
1305-B	Zone 5 (Campbellton area)
1306-B	Zone 6 (Bathurst area)
1307-B	Zone 7 (Miramichi area)
24	Quebec
2401-F	Région du Bas-Saint-Laurent
2402-F	Région du Saguenay – Lac-Saint-Jean
2403-F	Région de la Capitale-Nationale
2404-F	Région de la Mauricie et du Centre-du-Québec
2405-F	Région de l'Estrie
2406-C	Région de Montréal
2407-G	Région de l'Outaouais
2408-F	Région de l'Abitibi-Témiscamingue
2409-F	Région de la Côte-Nord
2410-F	Région du Nord-du-Québec
2411-B	Région de la Gaspésie – Îles-de-la-Madeleine
2412-F	Région de la Chaudière-Appalaches
2413-G	Région de Laval
2414-F	Région de Lanaudière
2415-I	Région des Laurentides
2416-F	Région de la Montérégie
2417-A	Région du Nunavik
2418-A	Région des Terres-Cries-de-la-Baie-James

35	Ontario by Health Regions
3501-C	West
3502-B	Central
3503-B	Toronto
3504-B	East
3505-H	North East
3506-H	North West
35	Ontario by Health Units
3526-B	The District of Algoma Health Unit
3527-I	Brant County Health Unit
3530-H	Durham Regional Health Unit
3533-I	Grey Bruce Health Unit
3534-I	Haldimand-Norfolk Health Unit
3535-I	Haliburton, Kawartha, Pine Ridge District Health Unit
3536-D	Halton Regional Health Unit
3537-G	City of Hamilton Health Unit
3538-I	Hastings and Prince Edward Counties Health Unit
3540-I	Chatham-Kent Health Unit
3541-I	Kingston, Frontenac and Lennox and Addington Health Unit
3542-I	Lambton Health Unit
3543-I	Leeds, Grenville and Lanark District Health Unit
3544-G	Middlesex–London Health Unit
3546-I	Niagara Regional Area Health Unit
3547-I	North Bay Parry Sound District Health Unit
3549-E	Northwestern Health Unit
3550-I	Huron Perth Health Unit
3551-D	City of Ottawa Health Unit
3553-D	Peel Regional Health Unit
3555-I	Peterborough County–City Health Unit
3556-E	Porcupine Health Unit
3557-I	Renfrew County and District Health Unit
3558-I	The Eastern Ontario Health Unit
3560-I	Simcoe Muskoka District Health Unit
3561-I	Sudbury and District Health Unit
3562-E	Thunder Bay District Health Unit
3563-E	Timiskaming Health Unit
3565-G	Waterloo Health Unit
3566-H	Wellington–Dufferin-Guelph Health Unit
3568-G	Windsor–Essex County Health Unit
3570-D	York Regional Health Unit
3575-I	Oxford Elgin St. Thomas Health Unit
3595-C	City of Toronto Health Unit
46	Manitoba
4601-G	Winnipeg-Churchhill Health Region
4602-E	Prairie Mountain Health Region
4603-E	Interlake-Eastern Health Region
4604-A	Northern Health Region

4605-E	Southern Health Region
47	Saskatchewan
4721-A	Far North
4722-E	North Central West
4723-E	North Central East
4724-G	Saskatoon
4725-E	South West
4726-E	South East
4727-G	Regina
48	Alberta
4831-E	South Zone
4832-D	Calgary Zone
4833-E	Central Zone
4834-D	Edmonton Zone
4835-E	North Zone
59	British Columbia
5911-I	East Kootenay Health Service Delivery Area
5912-I	Kootenay-Boundary Health Service Delivery Area
5913-I	Okanagan Health Service Delivery Area
5914-I	Thompson/Cariboo Health Service Delivery Area
5921-H	Fraser East Health Service Delivery Area
5922-H	Fraser North Health Service Delivery Area
5923-D	Fraser South Health Service Delivery Area
5931-C	Richmond Health Service Delivery Area
5932-C	Vancouver Health Service Delivery Area
5933-H	North Shore/Coast Garibaldi Health Service Delivery Area
5941-H	South Vancouver Island Health Service Delivery Area
5942-I	Central Vancouver Island Health Service Delivery Area
5943-I	North Vancouver Island Health Service Delivery Area
5951-E	Northwest Health Service Delivery Area
5952-E	Northern Interior Health Service Delivery Area
5953-E	Northeast Health Service Delivery Area
60	Yukon
6001-E	Yukon
61	Northwest Territories
6101-E	Northwest Territories
62	Nunavut
6201-A	Nunavut

A	Peer group A
B	Peer group B
C	Peer group C
D	Peer group D
E	Peer group E
F	Peer group F

G	Peer group G
H	Peer group H

APPENDIX C - SAMPLE ALLOCATION BY PROVINCE (CCHS 2023)

Appendix C – Sample allocation by province (CCHS 2023)[11]

Province	CCHS		Total
	Area Frame	DDAP	
Total	100,375	32,838	133,213
N.L.	2,978	1,518	4,496
P.E.I.	1,037	864	1,901
N.S.	3,873	2,191	6,064
N.B.	3,468	1,593	5,061
Que.	20,301	5,623	25,924
Ont.	30,839	6,207	37,046
Man.	5,003	3,026	8,029
Sask.	4,252	2,470	6,722
Alta.	13,130	4,634	17,764
B.C.	15,494	4,712	20,206

[11] These raw sample sizes correspond to the number of units that were sent to collection.

APPENDIX D - RESPONSE RATES BY PROVINCE (CCHS 2023)

Appendix D – Response rates by province (CCHS 2023)

Province	Area Frame			DDAP			Combined
	# in-scope	# respondents	Response rates	# in scope	# respondents	Response rates	Response rates
Total	95,281	38,952	40.9	32,051	12,688	39.6	40.6
N.L.	2,776	994	35.8	1,486	559	37.6	36.4
P.E.I.	919	313	34.1	831	368	44.3	38.9
N.S.	3,601	1,574	43.7	2,133	973	45.6	44.4
N.B.	3,192	1,175	36.8	1,544	726	47.0	40.1
Que.	19,060	8,335	43.7	5,433	2,330	42.9	43.5
Ont.	29,484	12,593	42.7	6,087	2,337	38.4	42.0
Man.	4,760	1,856	39.0	2,947	1,215	41.2	39.8
Sask.	4,046	1,370	33.9	2,414	899	37.2	35.1
Alta.	12,608	4,627	36.7	4,554	1,539	33.8	35.9
B.C.	14,835	6,115	41.2	4,622	1,742	37.7	40.4